

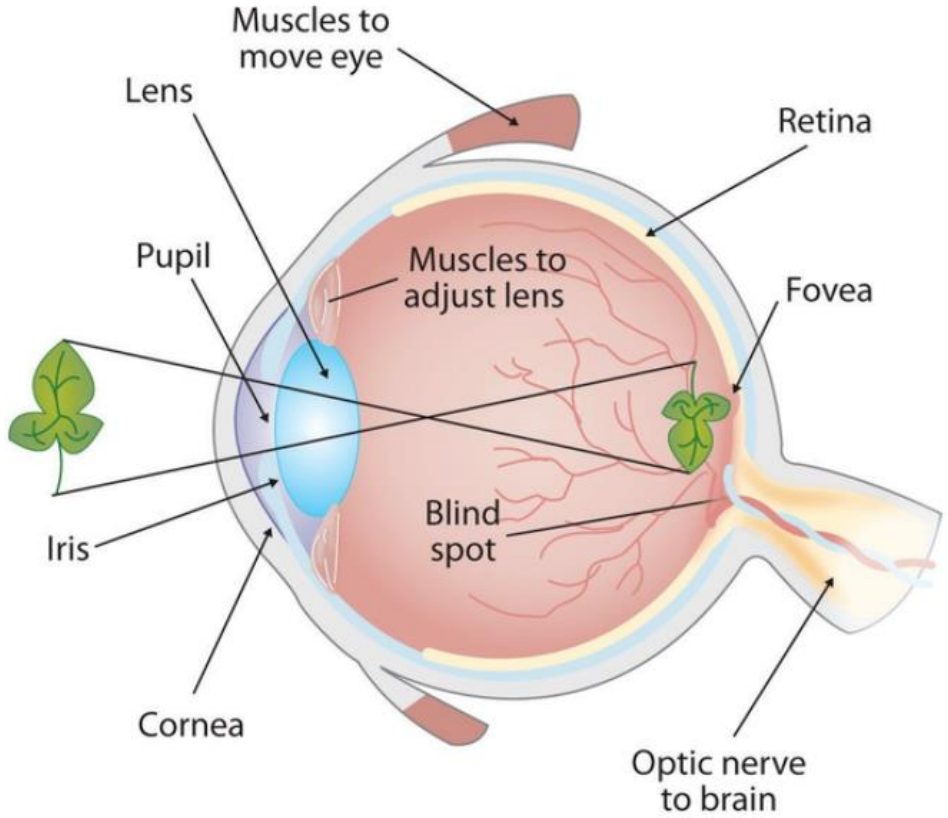




Human Visual Perception

Image Processing (310-3311)

Why do we have to study about our eyes in image processing class ?



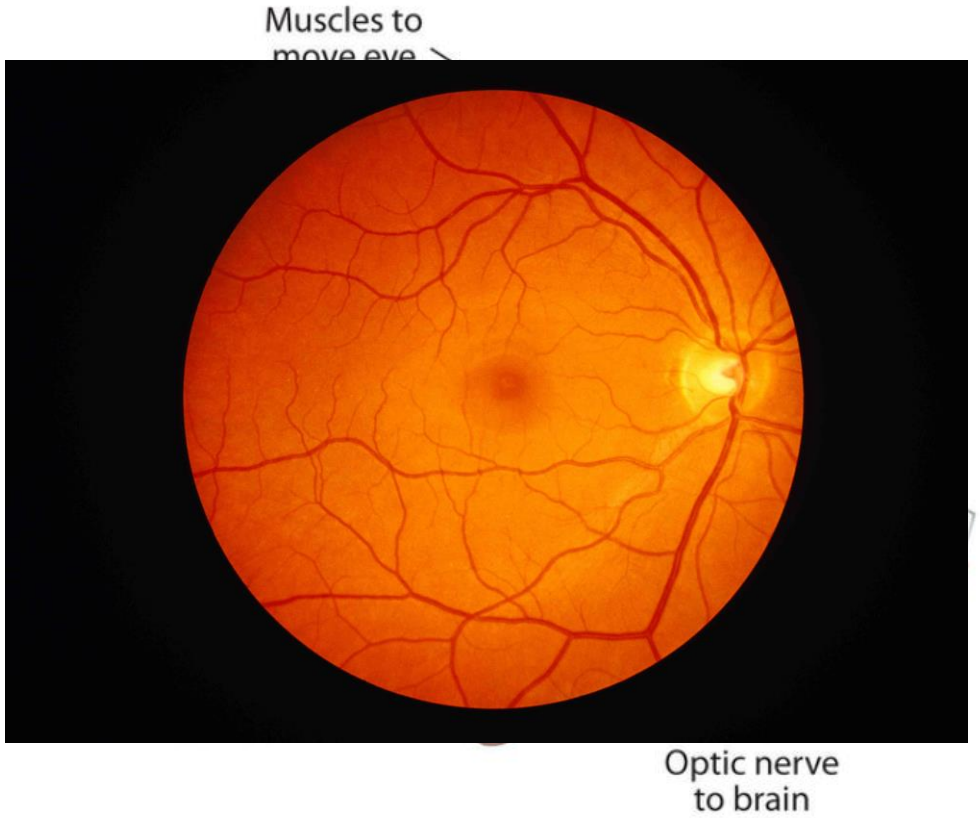
Cornea: A tough, transparent tissue that covers the anterior surface of the eye.

Lens: Makes up of concentric layers of fibrous cells and contains 60-70% of water. To focus on objects near or far the eye, the controlling muscles cause the lens to be thicker or thinner.

Retina: innermost membrane lines the inside of the wall's entire posterior portion. Retina contains **cones** and **rods**.

Ref: https://www.opentextbooks.org.hk/ditatopic/26428

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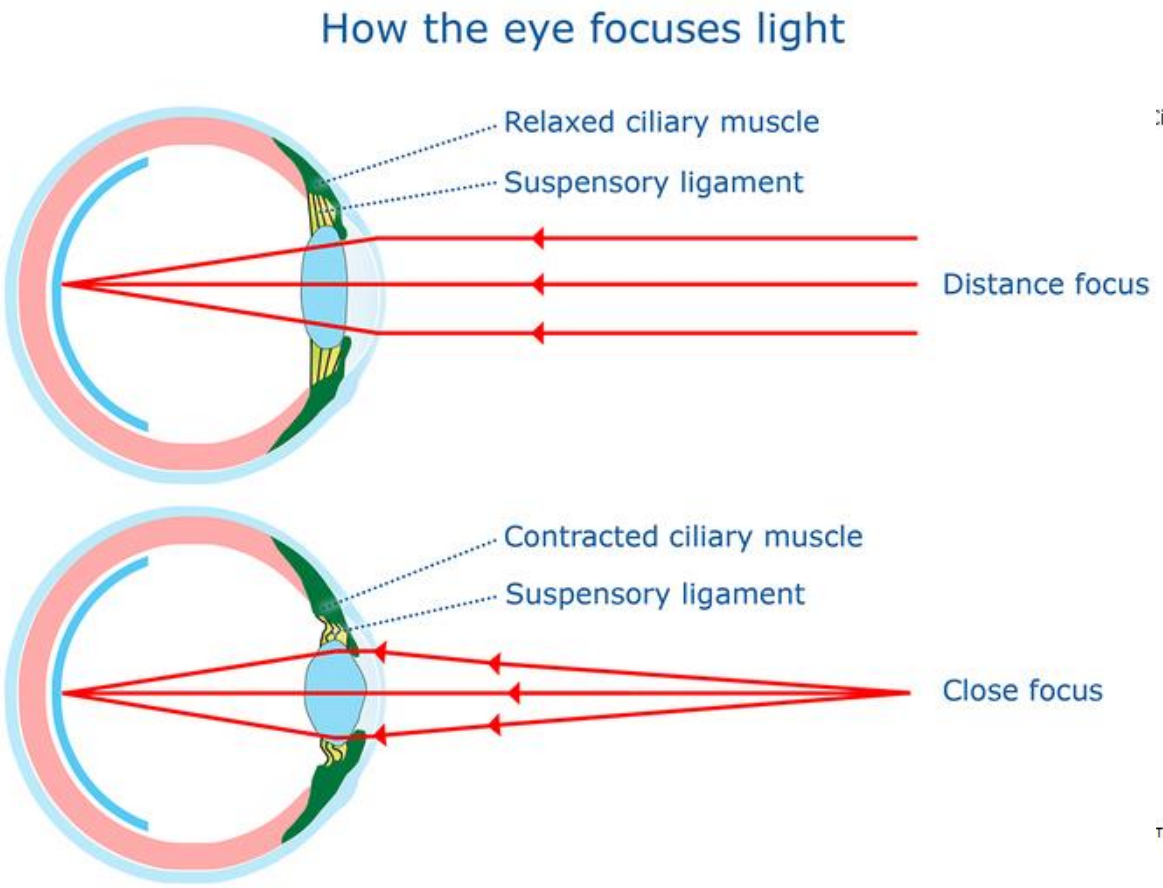
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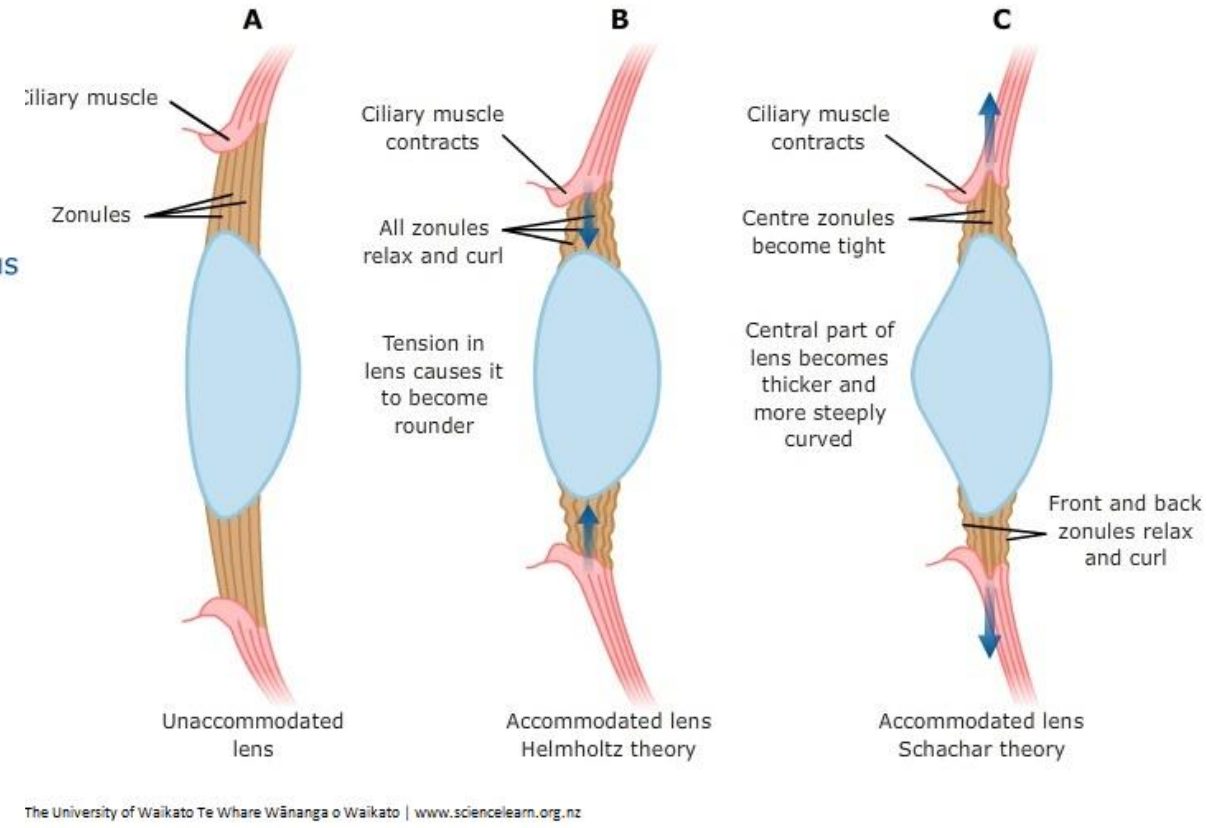
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Ref: https://www.opentextbooks.org.hk/ditatopic/26428

Comparing between lens of your eye and camera, what is big the difference ?

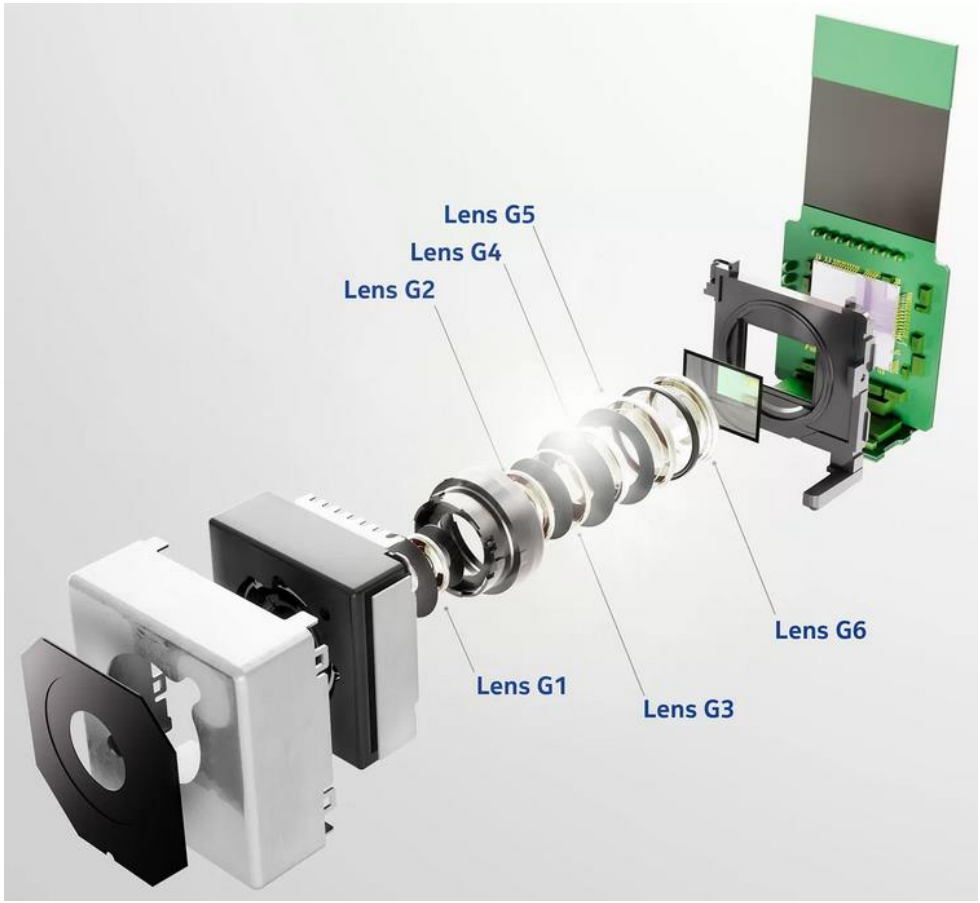




Ref: https://www.sciencelearn.org.nz/resources/50-how-the-eye-focuses-light

Comparing between lens of your eye and camera, what is big the difference ?

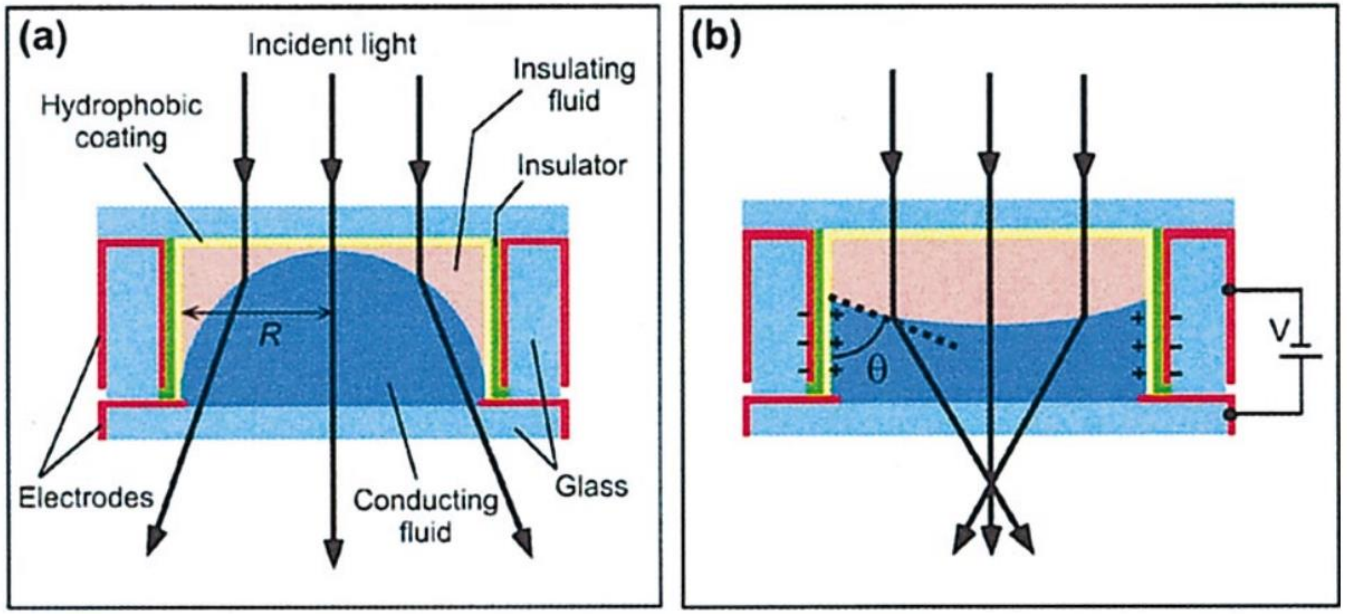




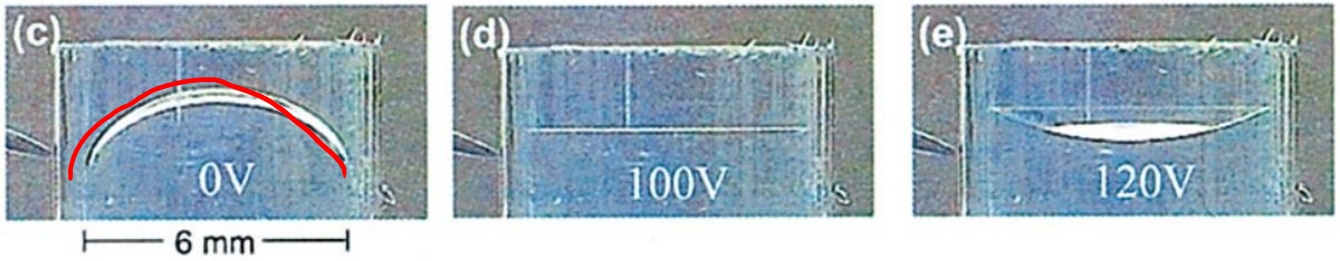
Smartphone camera

Ref: https://www.sciencelearn.org.nz/resources/50-how-the-eye-focuses-light

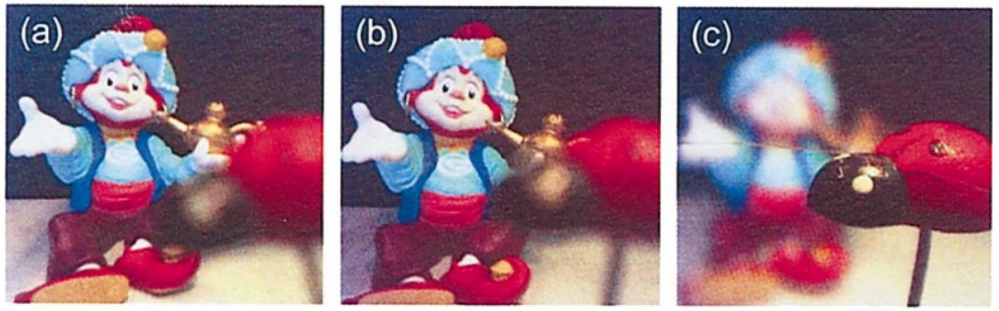
Why could we design our camera lens like our eyes?







Water & Oil



On Smartphone ?

Xiaomi Mi Mix Fold (available in China only)

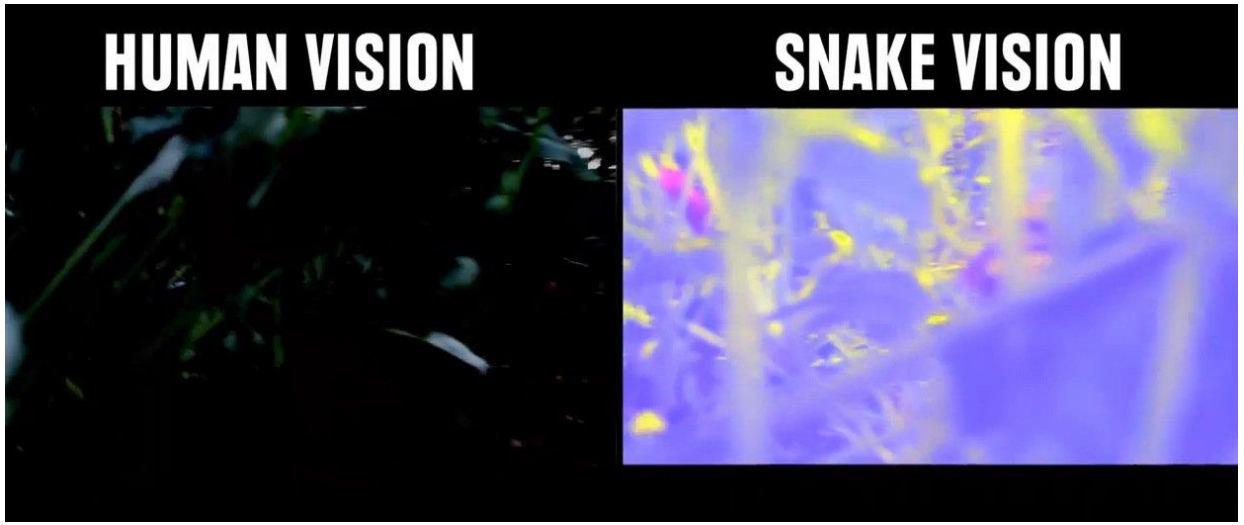


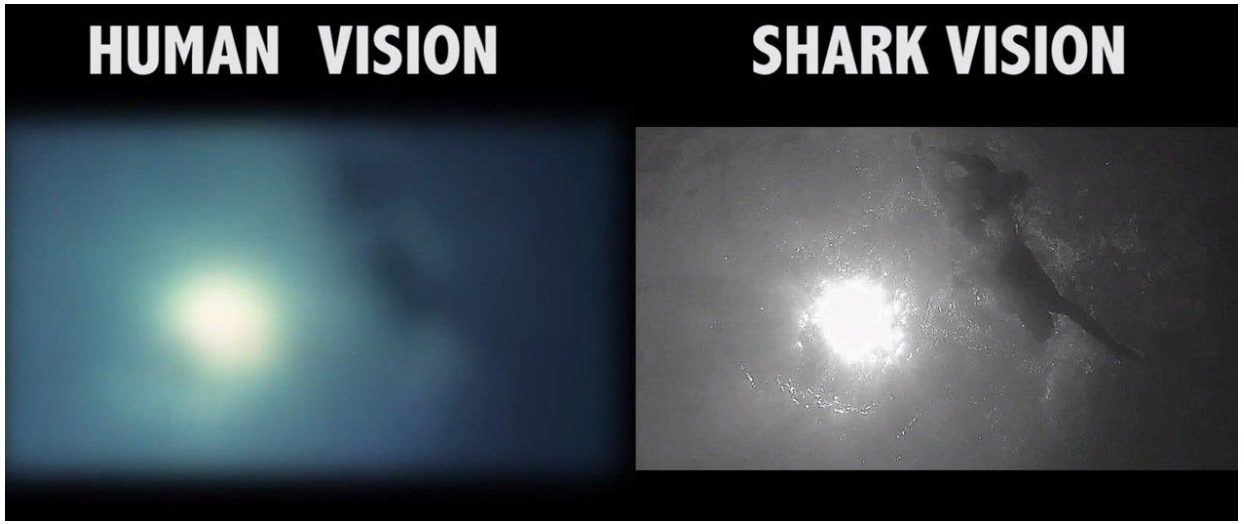
telephoto and macro by the same liquid lens

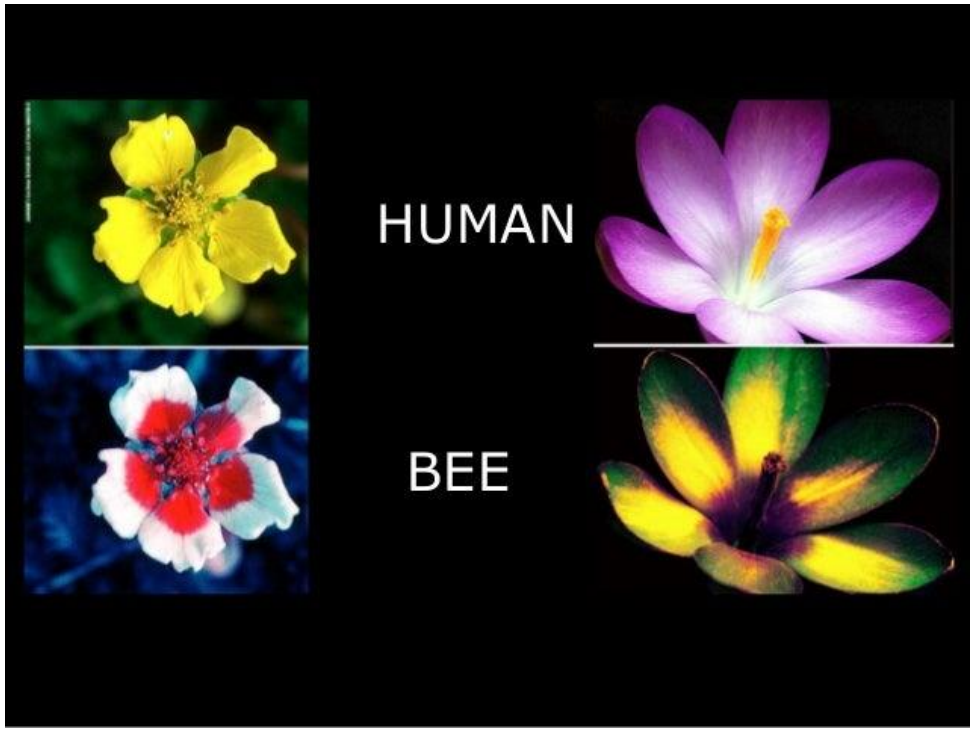


Using C1 professional imaging chip.
Mass production of mobile phone with liquid lenses.

Animals Eyes

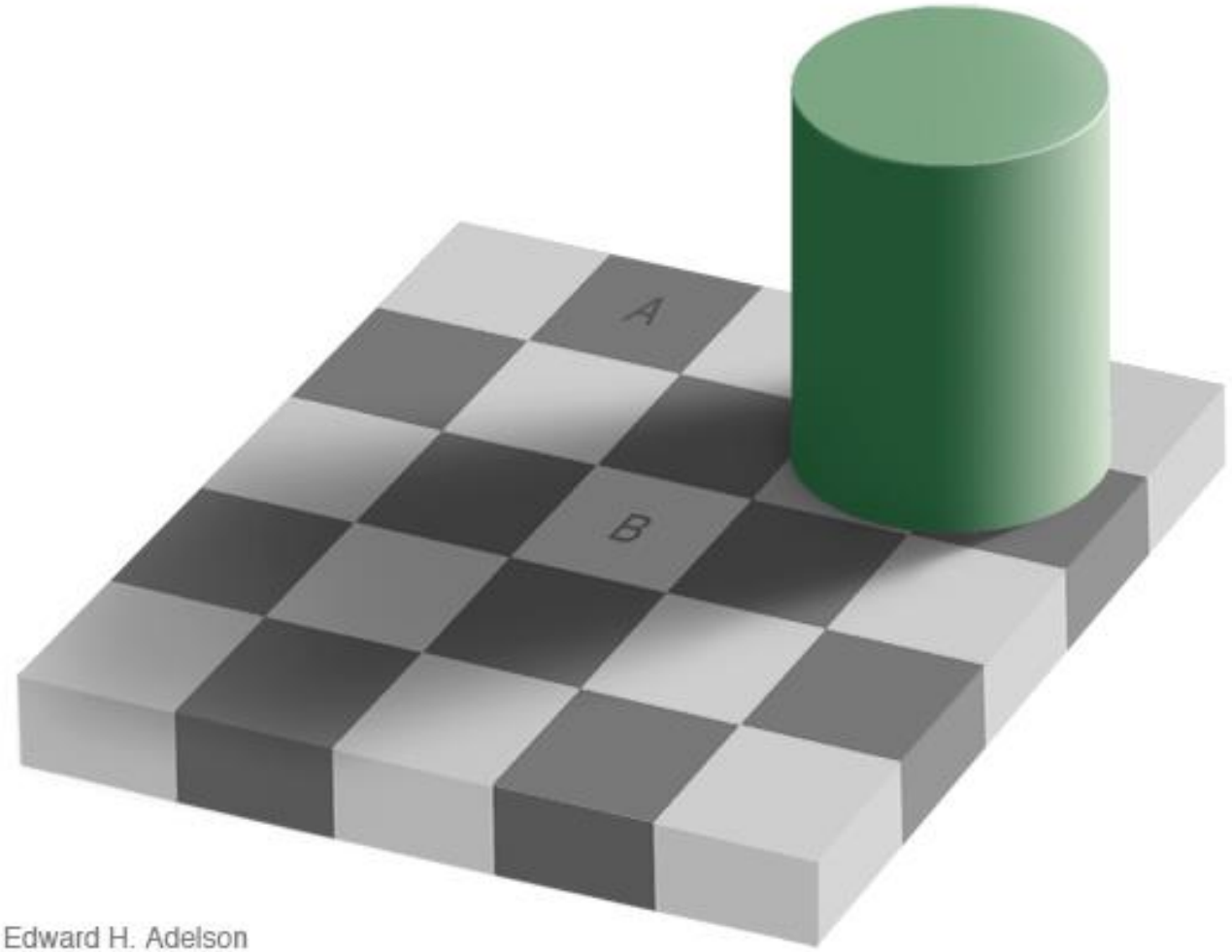




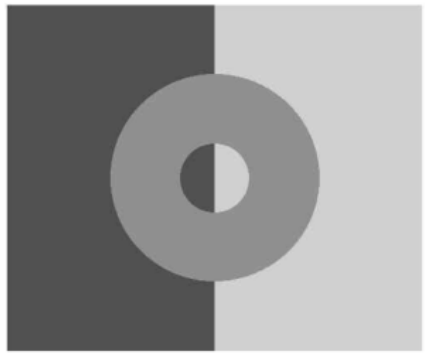


Ref: http://www.24hviralphotos.com/amazing-facts-about-human-vision-vs-animal-vision-8-pics

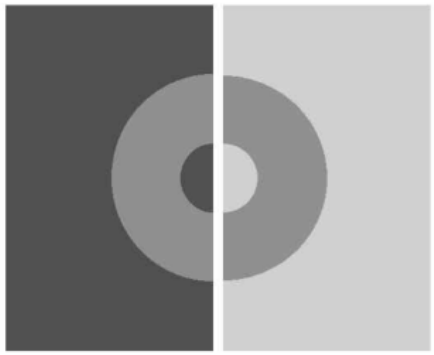
Can we trust our eyes?



Edward H. Adelson



a

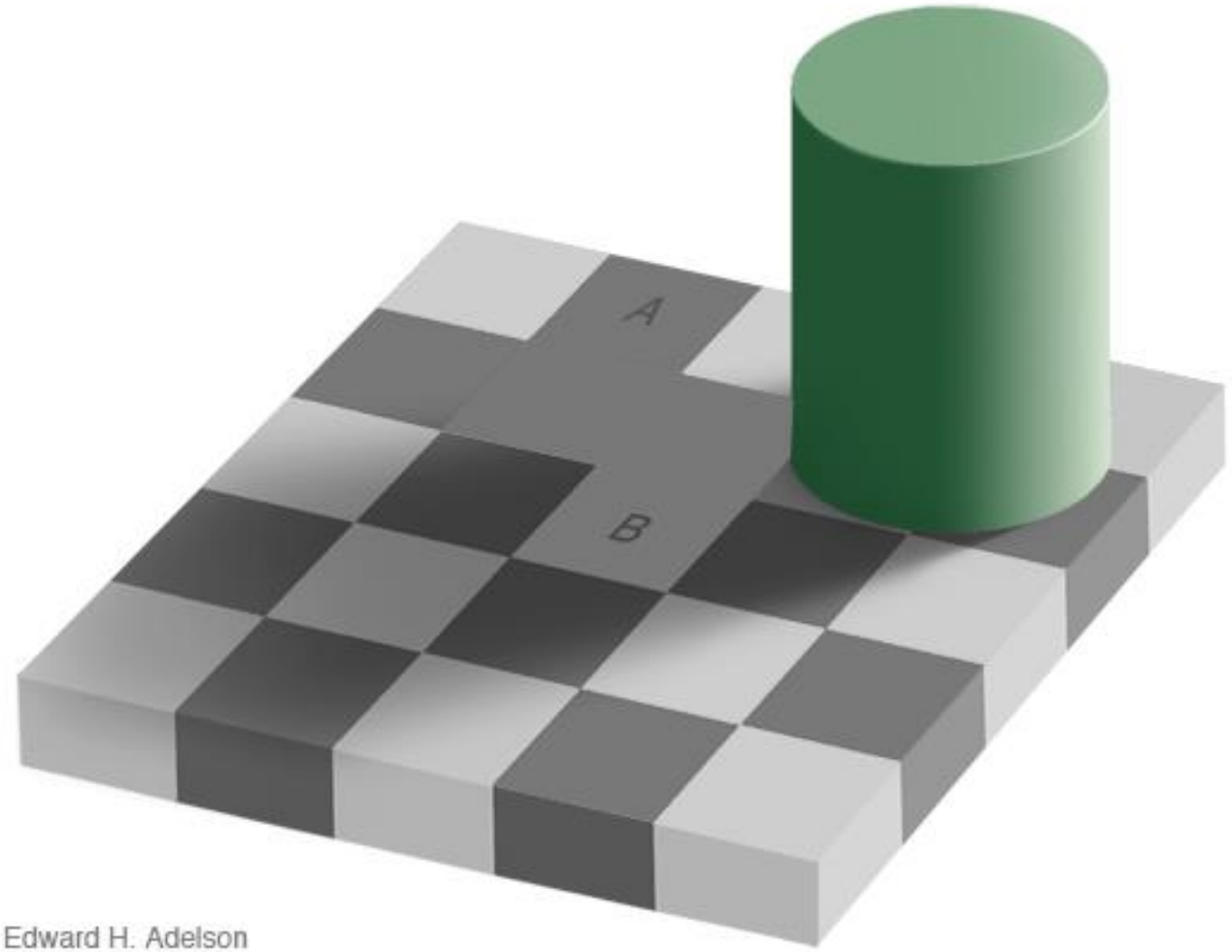


b

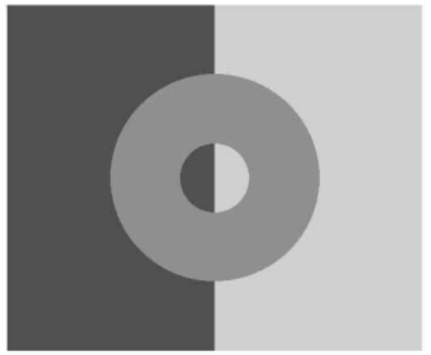
Koffka ring

Ref: http://persci.mit.edu/gallery

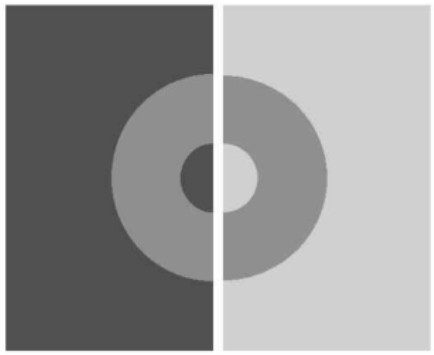
Can we trust our eyes?



Edward H. Adelson



a



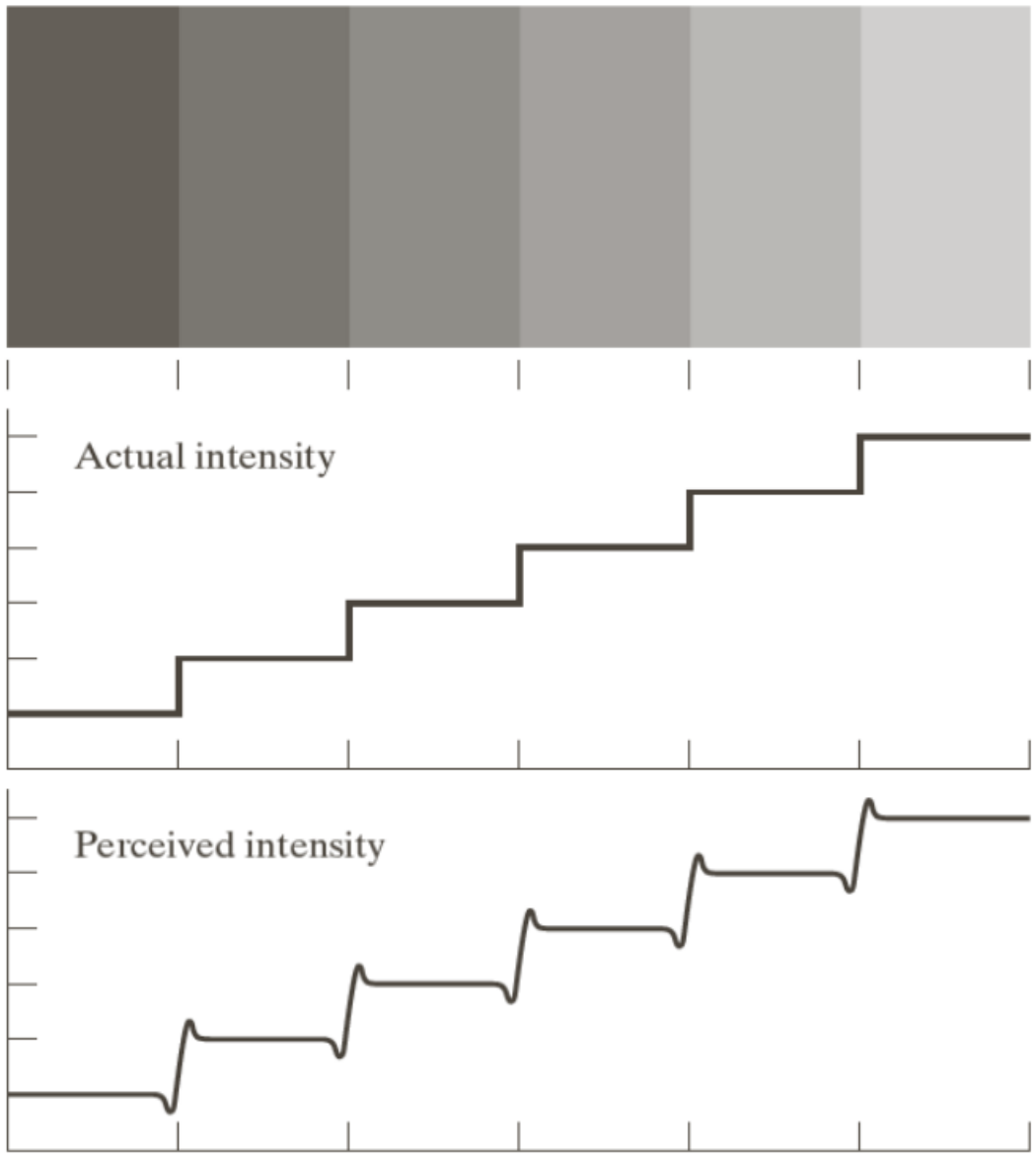
b

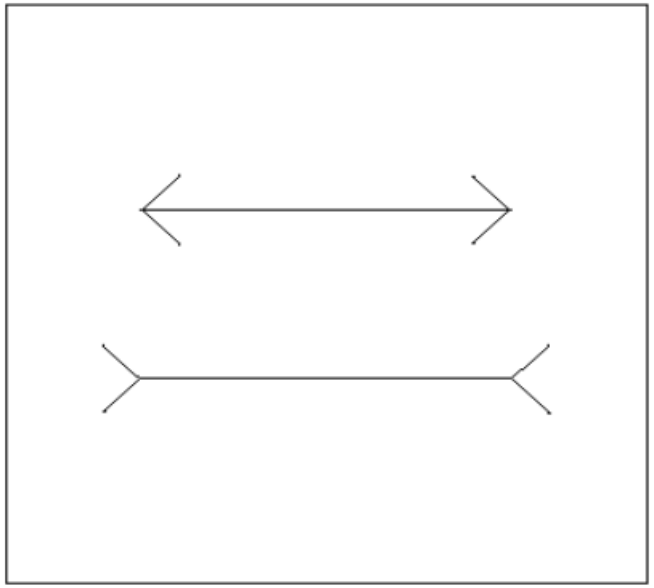


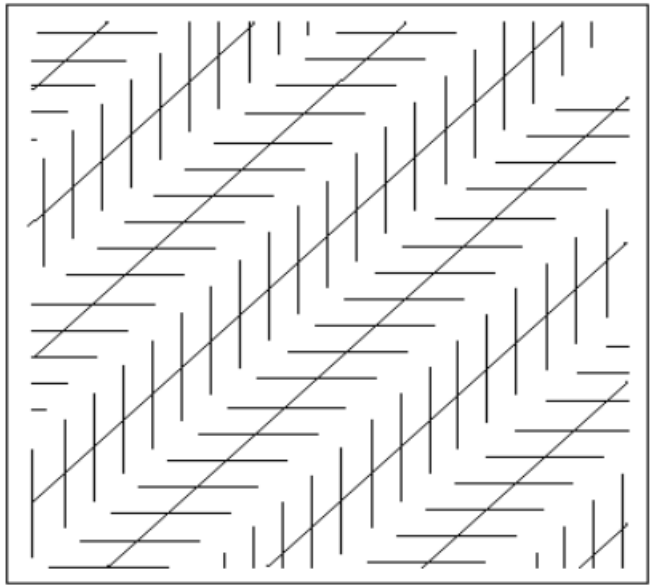
c

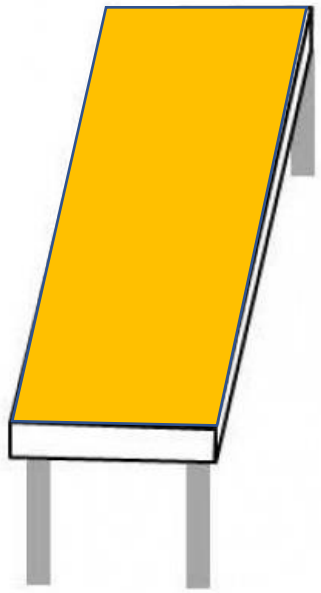
Koffka ring

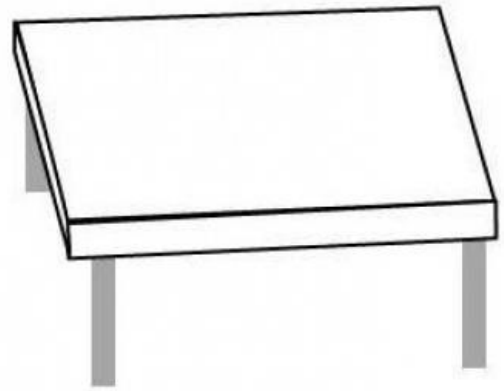
Ref: http://persci.mit.edu/gallery

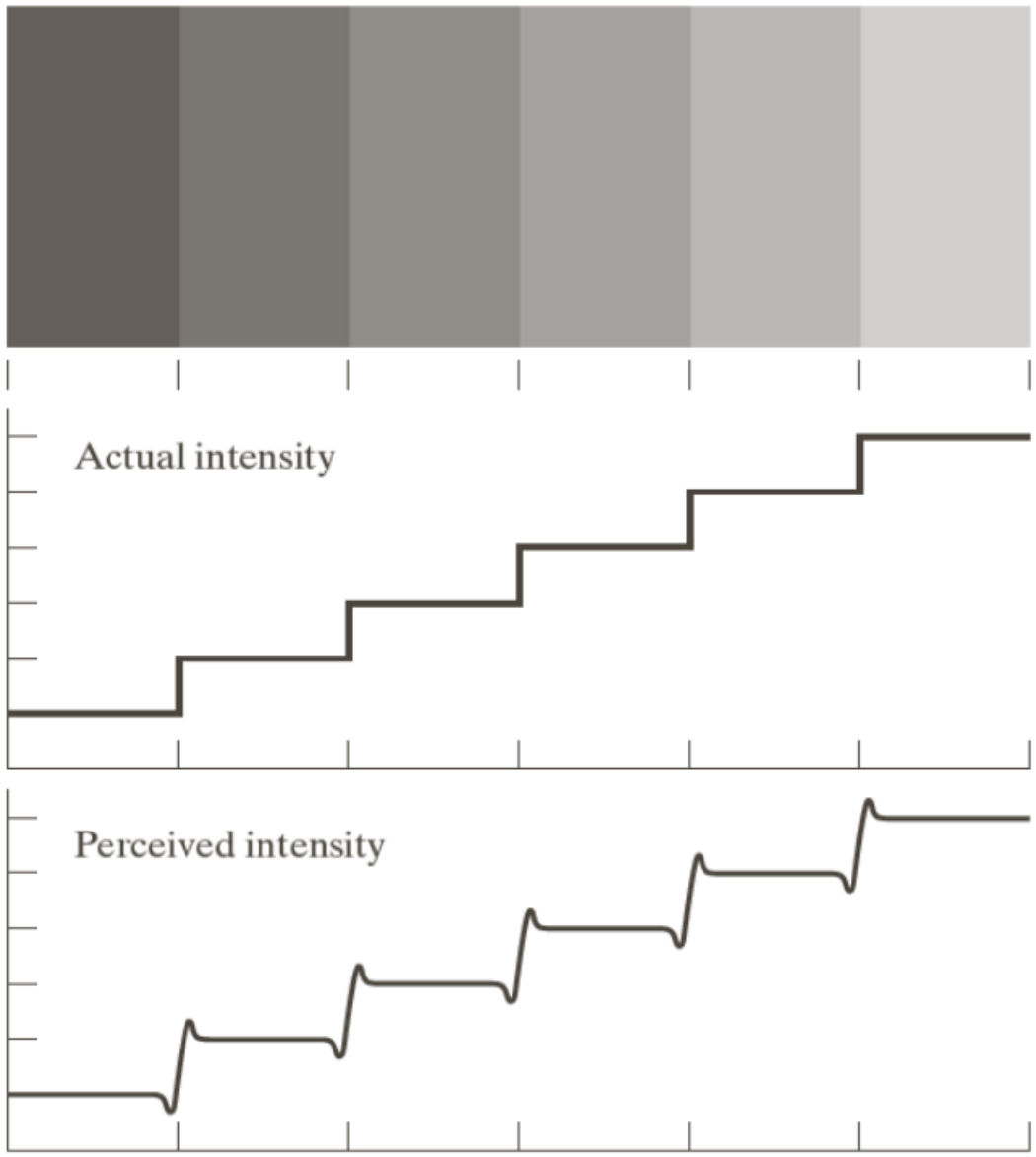

Actual intensity
Perceived intensity

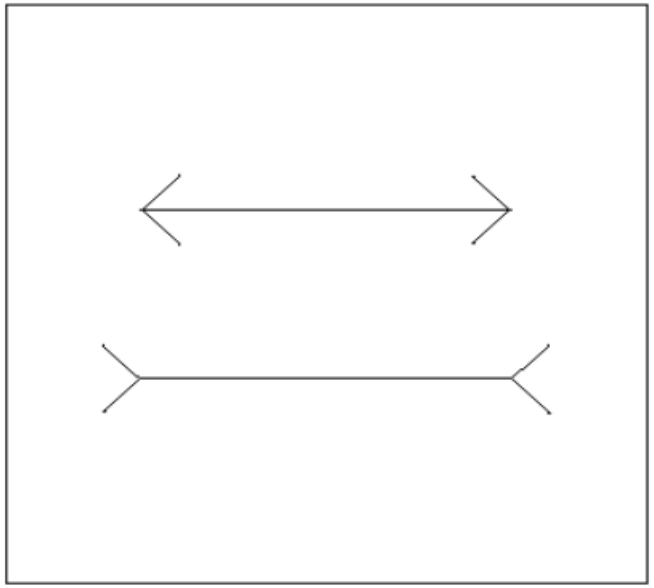


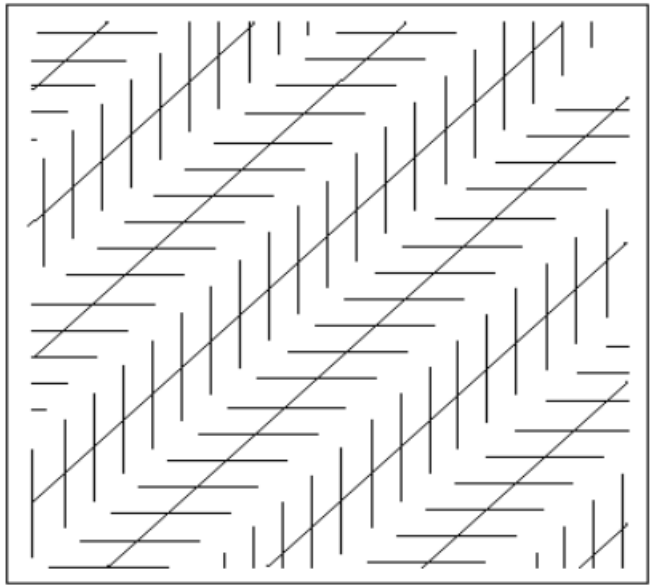


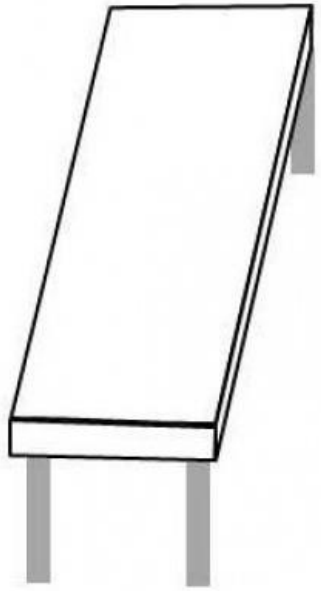


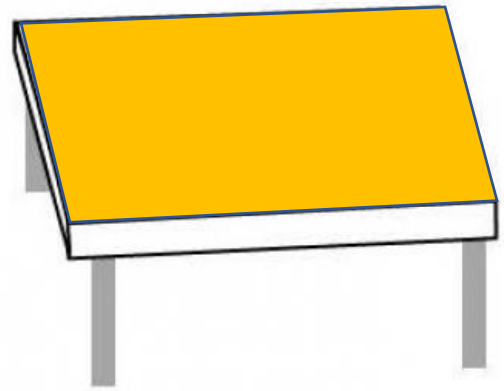



Actual intensity
Perceived intensity















Fundamental Operation

Image Processing (310-3311)

Digital Image (Grayscale)



We see “image” like this.

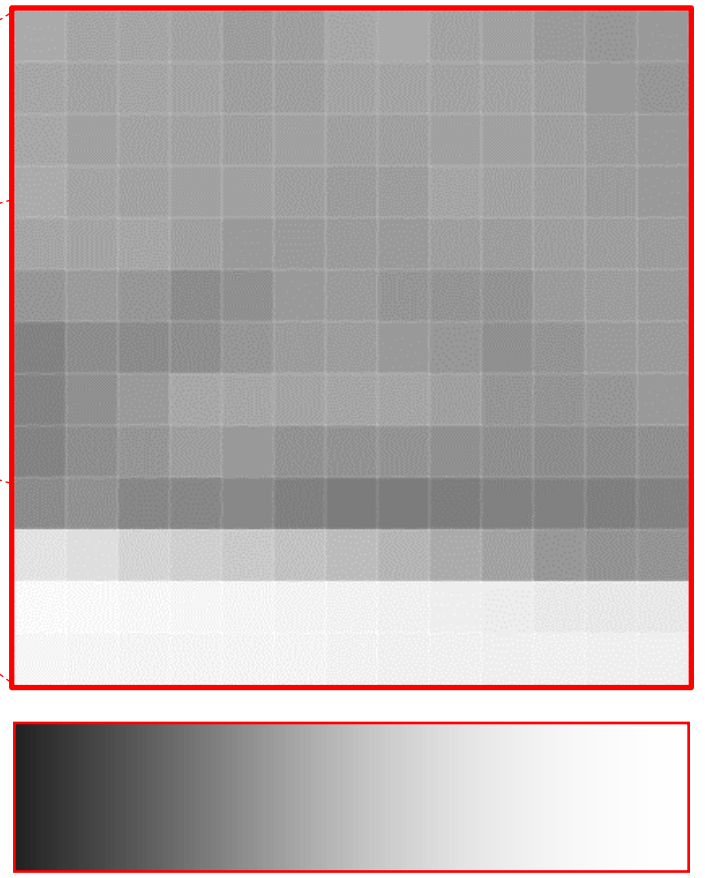
148	123	52	107	123	162	172	123	64	89	···
147	130	92	95	98	130	171	155	169	163	···
141	118	121	148	117	107	144	137	136	134	···
82	106	93	172	149	131	138	114	113	129	···
57	101	72	54	109	111	104	135	106	125	···
138	135	114	82	121	110	34	76	101	111	···
138	102	128	159	168	147	116	129	124	117	···
113	89	89	109	106	126	114	150	164	145	···
120	121	123	87	85	70	119	64	79	127	···
145	141	143	134	111	124	117	113	64	112	···
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	

But, this is what computer see.

Digital Image (Grayscale)



We see “image” like this.

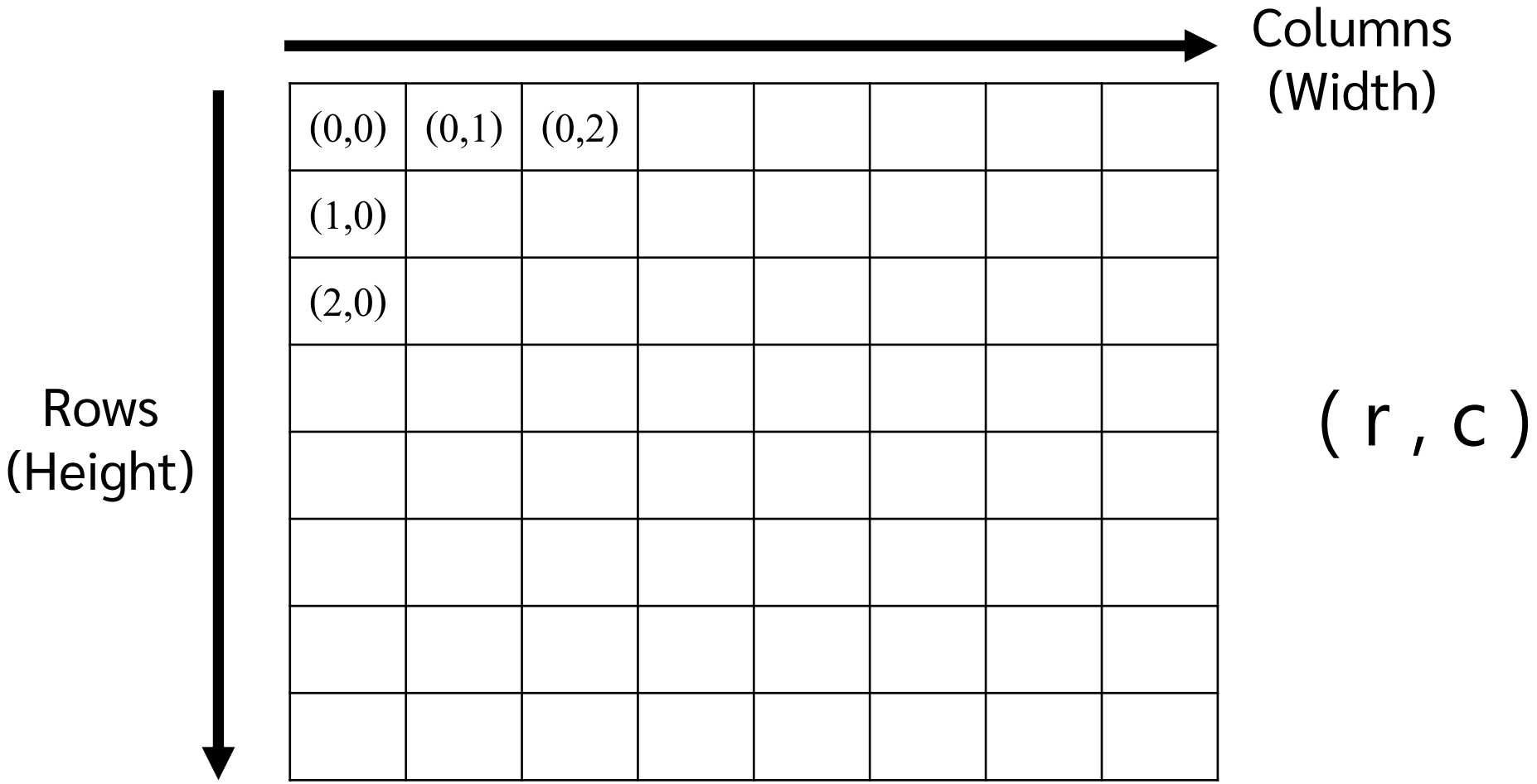


0 255

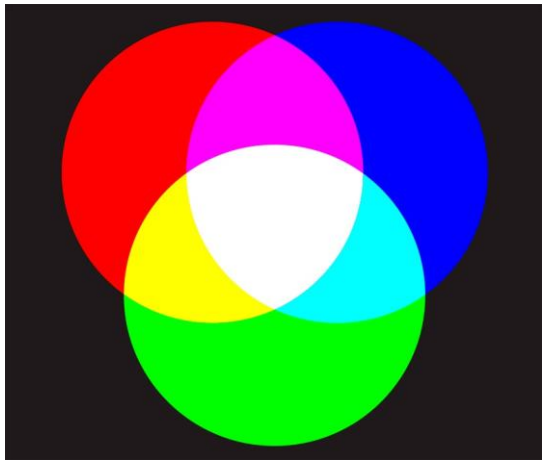
But, this is what computer see.

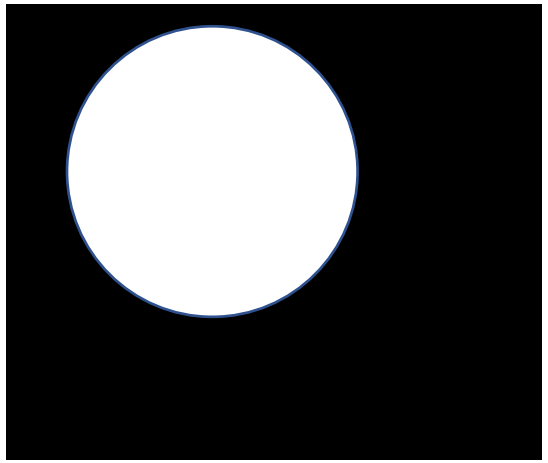
Digital Image ➔ Array (N-dimensions)

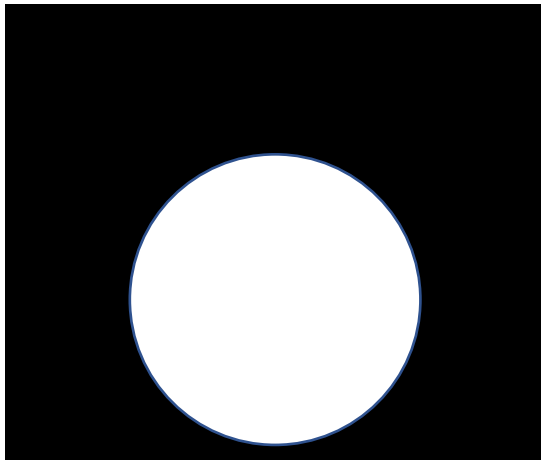
การจัดการ “ภาพ” ด้วย Python

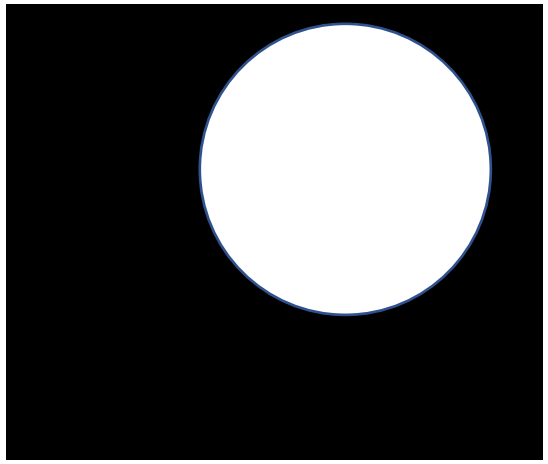


Digital Image (Color - RGB Model)

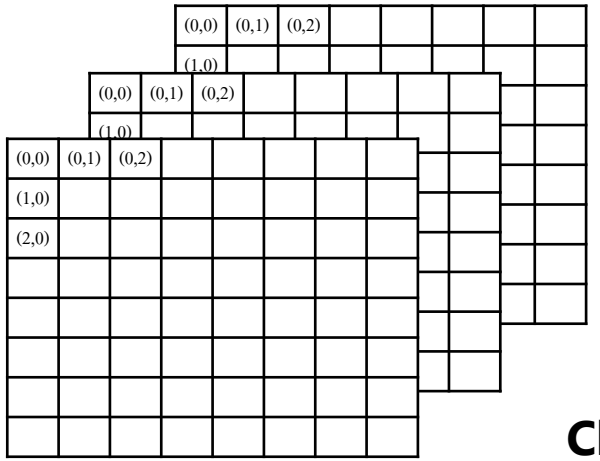








Red **Green** **Blue**



Channel : Red / Green / Blue

Python Implementation





1.) Check Python version

2.) Install PIP

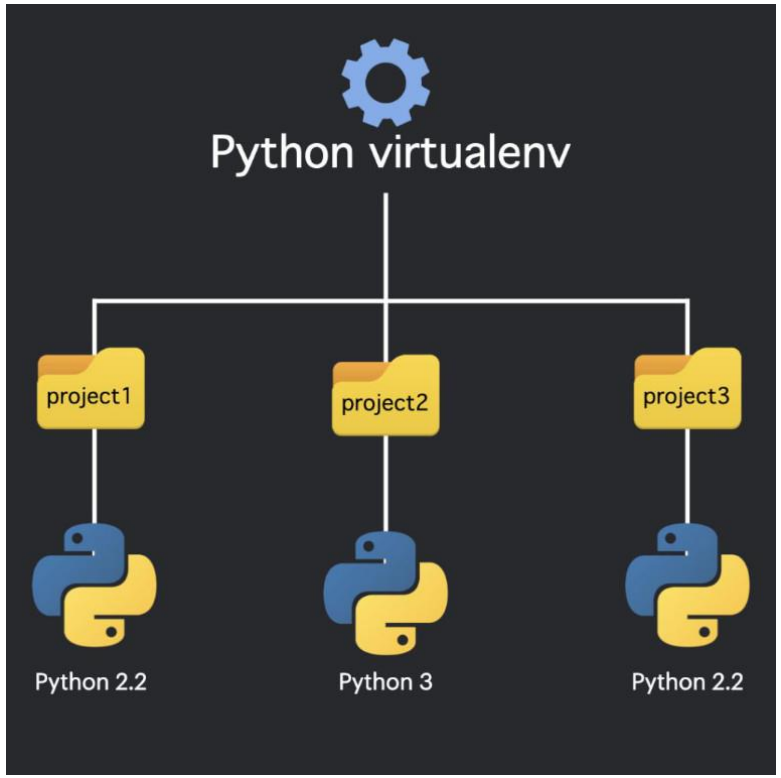
3.) Install library NumPy & Open CV

***** How to install PIP in macOS**

https://www.geeksforgeeks.org/how-to-install-pip-in-macos

Python Virtual Environment

“A virtual environment is a Python environment such that the Python interpreter, libraries and scripts installed into it are isolated from other”



1.) การสร้าง Virtual Environment

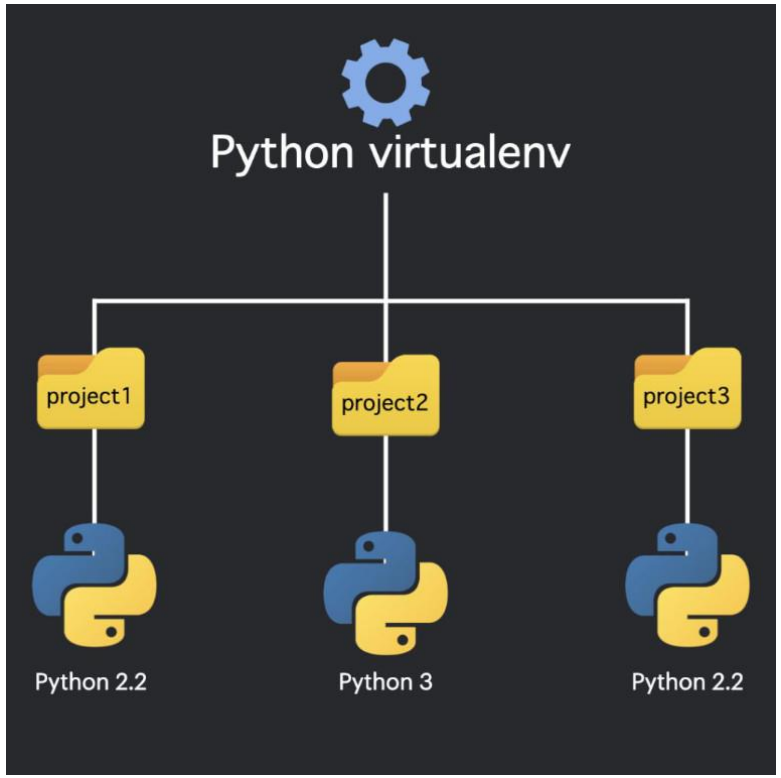
```
python -m venv <virtual-environment-name>
```

2.) การ Activate

```
[Run]   env/Scripts/activate.bat
```

Python Virtual Environment

“A virtual environment is a Python environment such that the Python interpreter, libraries and scripts installed into it are isolated from other”



3.) การสร้าง File requirements.txt สำหรับเก็บ List

pip freeze > requirements.txt

4.) การ Install Package จาก requirements.txt

pip install -r requirements.txt

Other recommended management approaches

1. ใช้ `pyenv` + `pyenv-virtualenv` (สำหรับจัดการหลาย Python versions + env)

- `pyenv` ช่วยให้ติดตั้ง Python หลายเวอร์ชันในเครื่องเดียวกัน เช่น Python 3.8, 3.10, 3.12 ฯลฯ
- `pyenv-virtualenv` ช่วยสร้าง virtual environment แต่ละโปรเจกต์ แยก Python version และ packages ได้ชัดเจน
- รองรับ Mac/Linux ดีมาก และ Windows (ผ่าน WSL หรือ Git Bash)

bash Copy Edit

```bash
# ตัวอย่างการติดตั้ง Python 3.10
pyenv install 3.10.13

# สร้าง environment
pyenv virtualenv 3.10.13 myproject-env

# ใช้ environment นี้
pyenv activate myproject-env
```

Other recommended management approaches

2. ใช้ `conda` หรือ `mamba` (เหมาะกับ Data Science / Windows user)

- ใช้ Anaconda หรือ Miniconda เพื่อจัดการ environment + Python version
- `mamba` เป็นเวอร์ชันเร็วกว่า `conda` (แนะนำ)
- เหมาะกับคนที่ทำงานกับพวก NumPy, SciPy, TensorFlow, etc.

bash Copy Edit

```bash
# สร้าง environment ใหม่พร้อม Python 3.9
conda create -n myenv python=3.9

# เข้า environment
conda activate myenv

# ติดตั้ง library ตามโปรเจกต์
conda install numpy pandas
```

Read / Write / Show image

Example #1

Demo02_Read-Write-Show.py - C:\Users\Krisada\...

File Edit Format Run Options Window Help

```python
import cv2 as cv

# Read the image
img = cv.imread("pic/B.bmp")

print("Data type of image (OpenCV): ", type(img))

# Display the image
cv.imshow("Example1 - imshow", img)

# Save the image
cv.imwrite("out.png", img)

# Wait for a key press
cv.waitKey(0)

# Clean up
cv.destroyAllWindows()

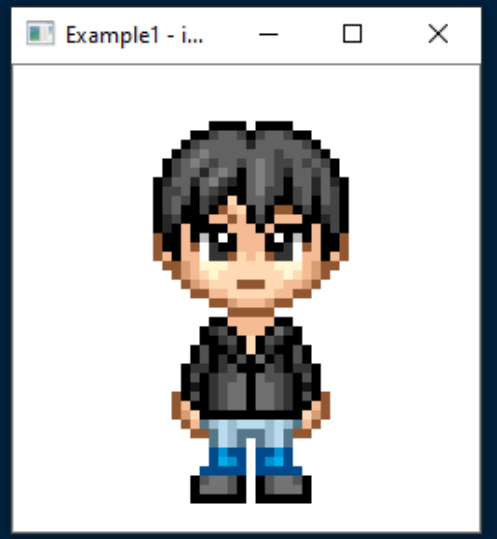
```

Ln: 19 Col: 0

IDLE Shell 3.9.1

File Edit Shell Debug Options Window Help

```
Python 3.9.1 (tags/v3.9.1:1e5d33e, Dec  7 2020, 16:33:24) [MSC v.1928 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Krisada\Desktop\DIP 2020\Code\Python\Demo 02\Demo02_Read-Write-Show.py
Data type of image (OpenCV):  <class 'numpy.ndarray'>
```



Ln: 5 Col: 0

Read / Write / Show image

Example #1

Demo02_Read-Write-Show.py - C:\Users\Krisada\...

File Edit Format Run Options Window Help

```
1  import cv2 as cv
2
3  # Read the image
4  img = cv.imread("pic/B.bmp")
5
6  print("Data type of image (OpenCV): ", type(img))
7
8  # Display the image
9  cv.imshow("Example1 - imshow", img)
10
11 # Save the image
12 cv.imwrite("out.png", img)
13
14 # Wait for a key press
15 cv.waitKey(0)
16
17 # Clean up
18 cv.destroyAllWindows()
19
```

Ln: 19 Col: 0

อ่านไฟล์ภาพที่อยู่ใน Path ที่กำหนด เก็บไว้ในตัวแปร [img]

แสดงภาพในตัวแปร [img] ออกมาทางหน้าต่าง

Save ภาพลงไปตาม Path ที่กำหนด

ให้รอรับการกดปุ่มอะไรก็ได้จาก Keyboard (ถ้าไม่ใส่หน้าต่างจะปิดลงเอง)

หลังจากกดปุ่มแล้วจึงสั่งให้ปิดหน้าต่างทั้งหมด

Access image attributes

Example #2

Demo02_02_Grayscale.py - C:\Users\Krisada\Desktop\DIP 2020\Cod...

File Edit Format Run Options Window Help

```
1  import cv2 as cv
2
3  # Read the image
4  img = cv.imread("pic/C.jpg", cv.IMREAD_GRAYSCALE)
5
6  print("Data type of image (OpenCV): ", type(img))
7  print("number of dimension: ", img.ndim)
8  print("size of each dimension: ", img.shape)
9  print("image height: ", img.shape[0])
10 print("image width: ", img.shape[1])
11
12 # Display the image
13 cv.namedWindow('Example1 - imshow', cv.WINDOW_NORMAL)
14 cv.imshow('Example1 - imshow',img)
15
16 # Wait for a key press
17 cv.waitKey(0)
18
19 # Clean up
20 cv.destroyAllWindows()
21
```

Ln: 21 Col: 0

IDLE Shell 3.9.1

File Edit Shell Debug Options Window Help

```
Python 3.9.1 (tags/v3.9.1:1e5d33e, Dec  7 2020, 16:33:24) [MSC v.1928 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Krisada\Desktop\DIP 2020\Code\Python\Demo 02\Demo02_02_Grayscale.
py
Data type of image (OpenCV):  <class 'numpy.ndarray'>
number of dimension:  2
size of each dimension:  (1440, 1080)
image height:  1440
image width:  1080
```

Ln: 10 Col: 0

ribu

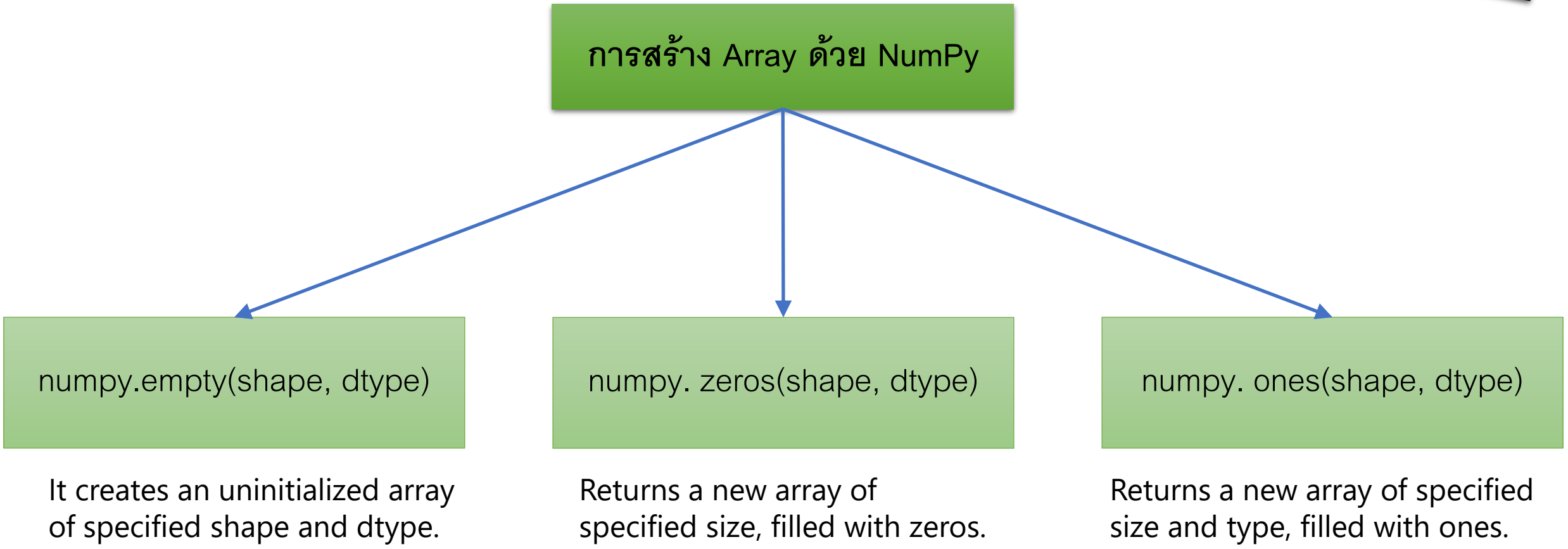
ole #2


ร่ม
ไม่ได้ถูกสร้างขึ้นมาเพื่อให้ฝนหยุดตก แต่
มันถูกสร้างขึ้นมาเพื่อให้เราเดินต่อไปได้ท่ามกลางสายฝน
กำลังใจ
ก็คงไม่ต่างจากร่ม เพราะมันไม่สามารถหยุดปัญหาหรือ
อุปสรรคที่จะเข้ามาได้ แต่
มันเป็นสิ่งที่จะพาเราก้าวผ่านสิ่งต่างๆ
เพื่อนำเราไปสู่จุดหมายปลายทางแห่งความสำเร็จได้
Club To Happiness
www.facebook.com/ClubToHappiness


Example1 - imshow
ร่ม
ไม่ได้ถูกสร้างขึ้นมาเพื่อให้ฝนหยุดตก แต่
มันถูกสร้างขึ้นมาเพื่อให้เราเดินต่อไปได้ท่ามกลางสายฝน
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Club To Happiness

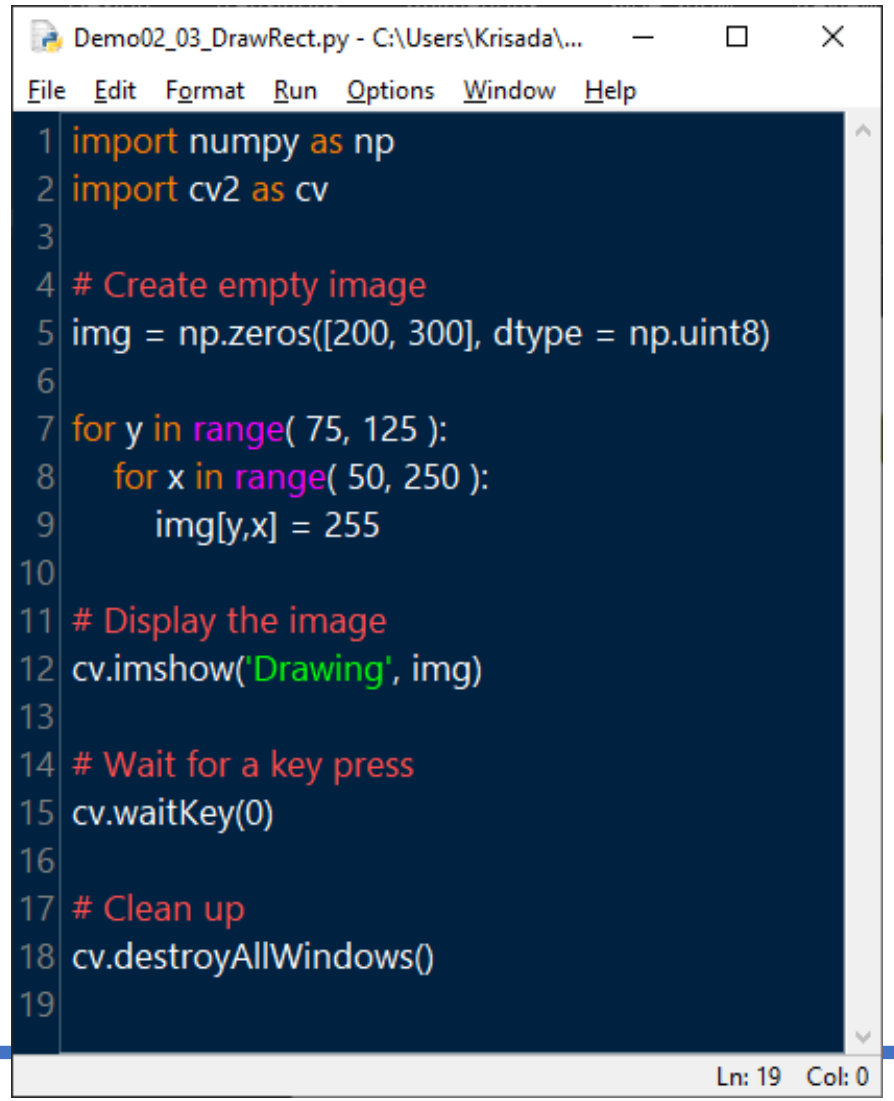
Access / Modify each pixel

Example #3



Access / Modify each pixel

Example #3



#	Data Types	Description
1	**bool_**	Boolean (True or False) stored as a byte
2	**int8**	Byte (-128 to 127)
3	**int16**	Integer (-32768 to 32767)
4	**int32**	Integer (-2147483648 to 2147483647)
5	**int64**	Integer (-9223372036854775808 to 9223372036854775807)
6	**uint8**	Unsigned integer (0 to 255)
7	**uint16**	Unsigned integer (0 to 65535)
8	**uint32**	Unsigned integer (0 to 4294967295)
9	**uint64**	Unsigned integer (0 to 18446744073709551615)
10	**float_**	Shorthand for float64
11	**float32**	Single precision float: sign bit, 8 bits exponent, 23 bits mantissa
12	**float64**	Double precision float: sign bit, 11 bits exponent, 52 bits mantissa
13	**complex**	Shorthand for complex128
14	**complex64**	Complex number, represented by two 32-bit floats (real and imaginary components)
15	**complex128**	Complex number, represented by two 64-bit floats (real and imaginary components)

Access / Modify each pixel

Example #3

Demo02_03_DrawRect.py - C:\Users\Krisada\...

File Edit Format Run Options Window Help

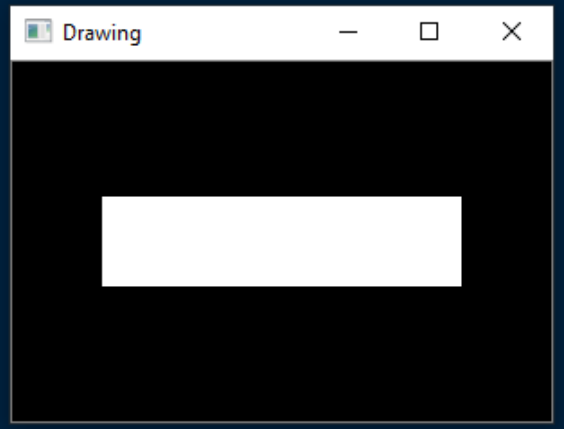
```
1  import numpy as np
2  import cv2 as cv
3
4  # Create empty image
5  img = np.zeros([200, 300], dtype = np.uint8)
6
7  for y in range( 75, 125 ):
8      for x in range( 50, 250 ):
9          img[y,x] = 255
10
11 # Display the image
12 cv.imshow('Drawing', img)
13
14 # Wait for a key press
15 cv.waitKey(0)
16
17 # Clean up
18 cv.destroyAllWindows()
19
```

Ln: 19 Col: 0

IDLE Shell 3.9.1

File Edit Shell Debug Options Window Help

```
Python 3.9.1 (tags/v3.9.1:1e5d33e, Dec  7 2020, 16:33:24) [MSC v.1928 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Krisada\Desktop\DIP 2020\Code\Python\Demo 02\Demo02_03_DrawRect.py
```



Ln: 5 Col: 0

What will happen if…

- จะเกิดอะไรขึ้นถ้ากำหนดให้ค่าในภาพมีค่าที่ไม่ได้อยู่ในช่วง 0 – 255 ???
- ถ้ามีการวนลูปออกนอกบริเวณภาพจะเกิดอะไรขึ้น

Try this…

- แยกภาพสี RGB ออกเป็น Plane สีย่อย Red, Green, Blue
- ทดลองเขียน Function ปรับค่าความสว่าง (Brightness) ของภาพ
- ทดลองใช้ Function **hconcat** หรือ **vconcat** เพื่อนำภาพมาวางต่อกัน

Image Acquisition (Real world → Image (Film))

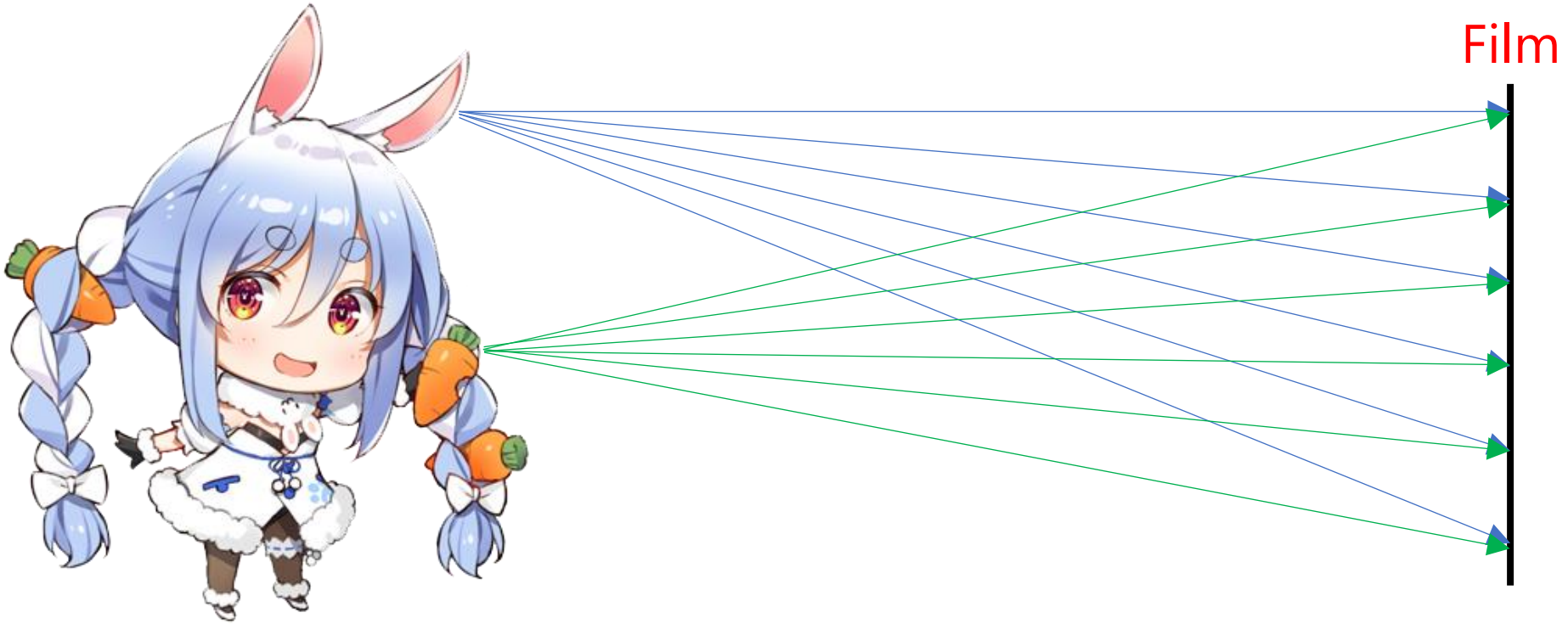


Image Acquisition (Pinhole Camera)

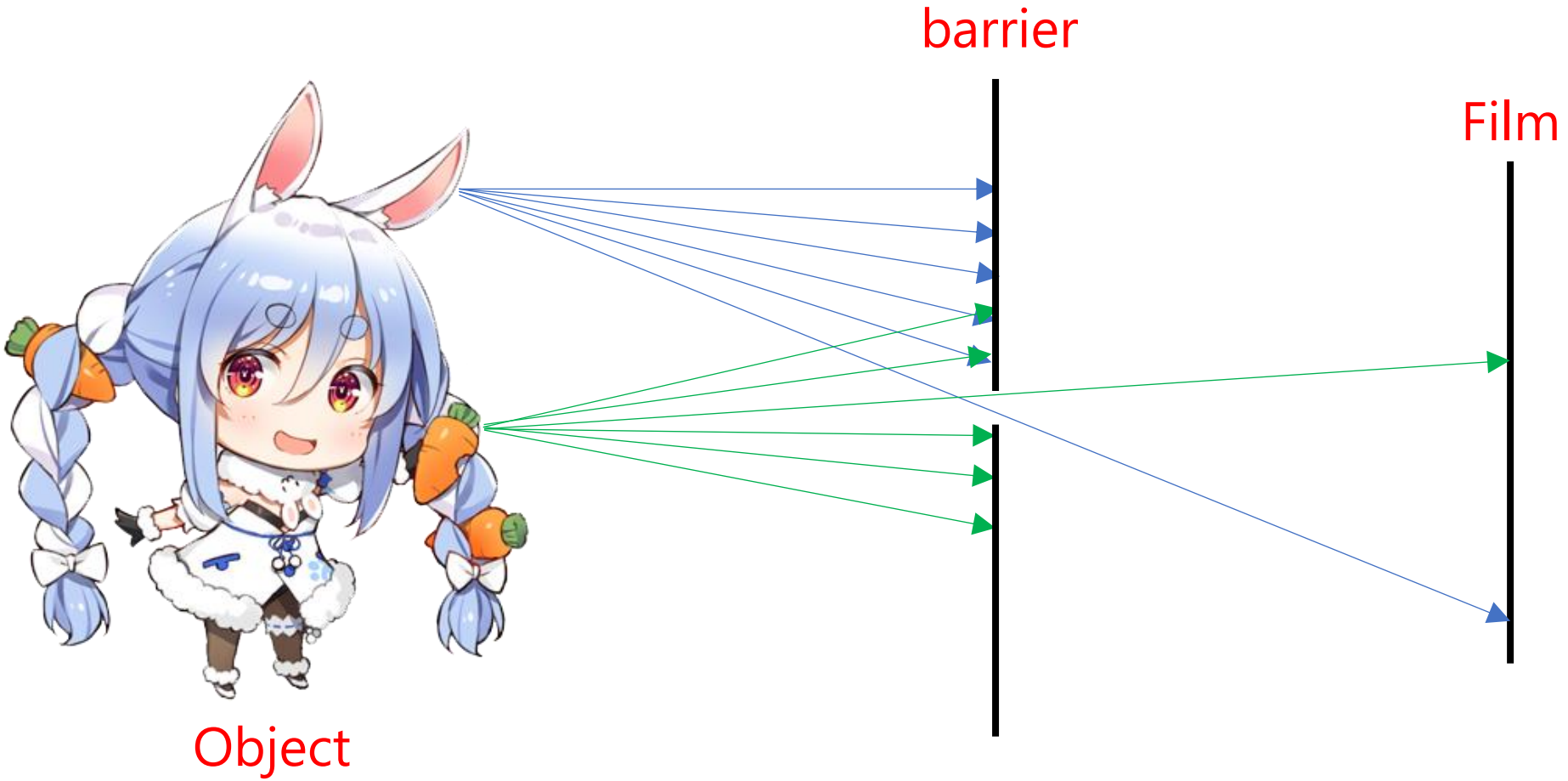
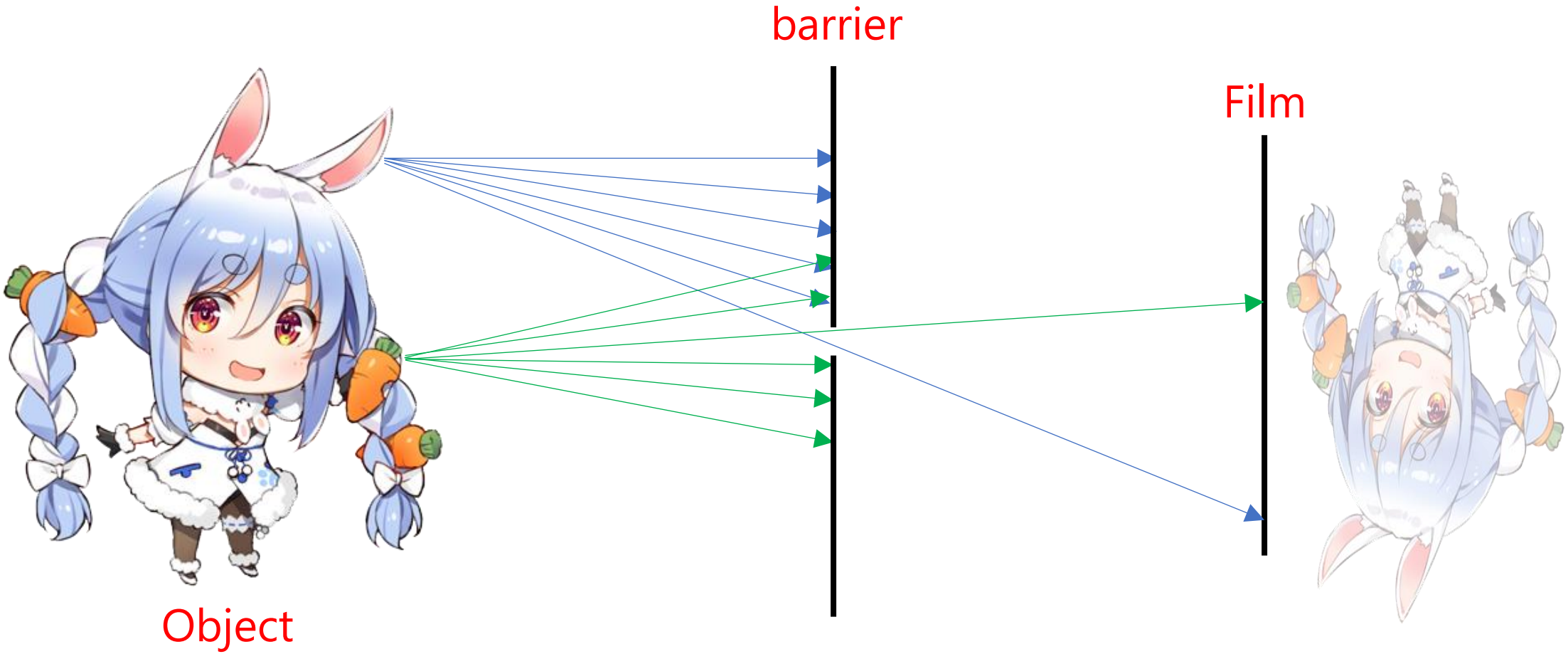
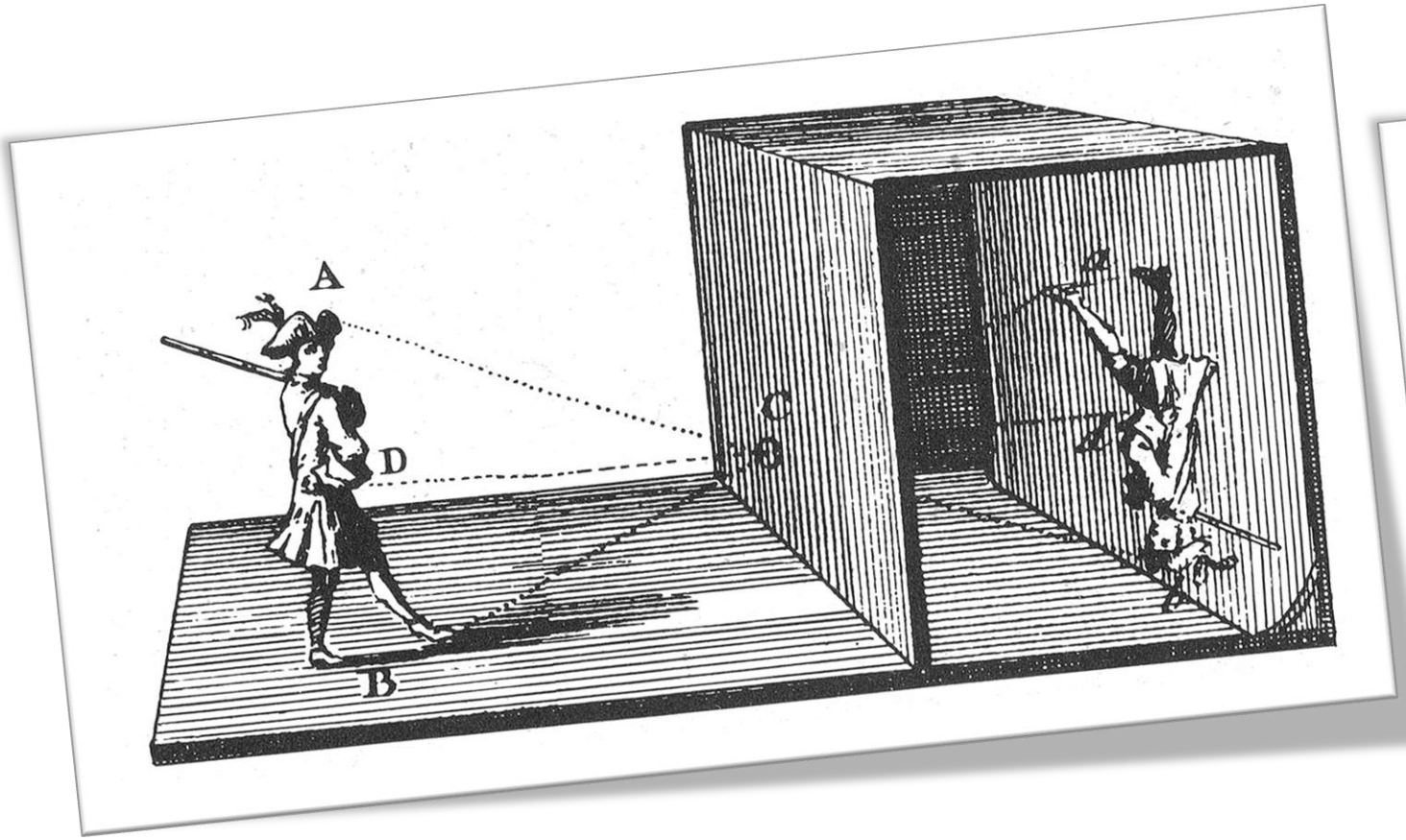
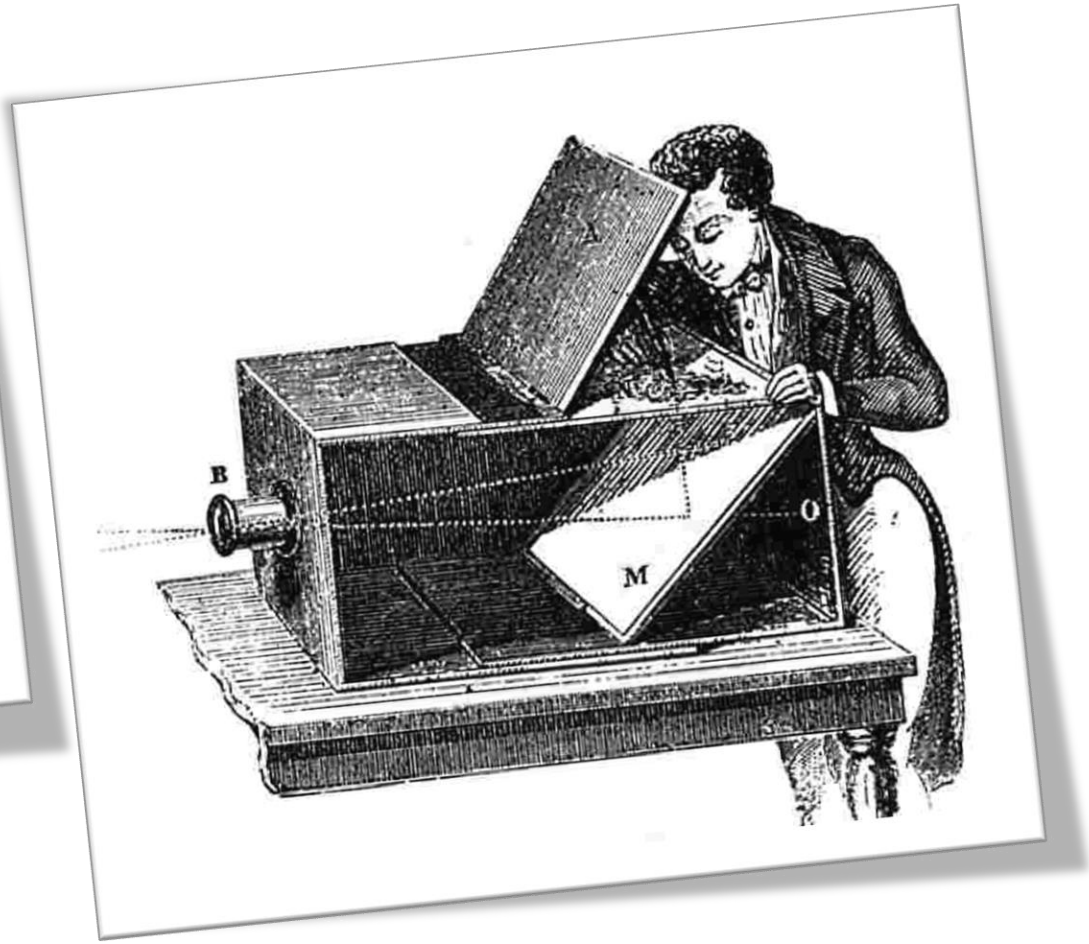


Image Acquisition (Pinhole Camera)

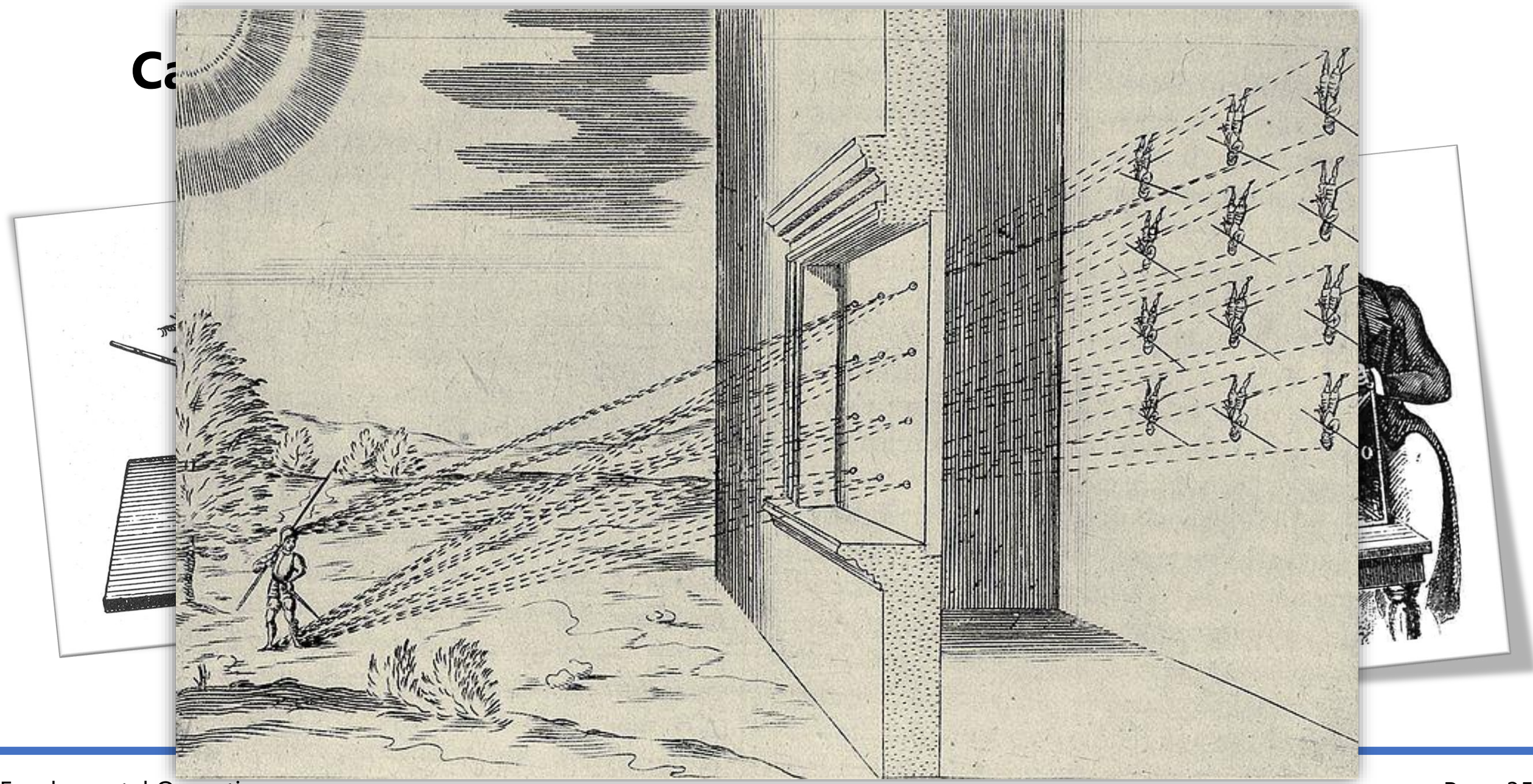


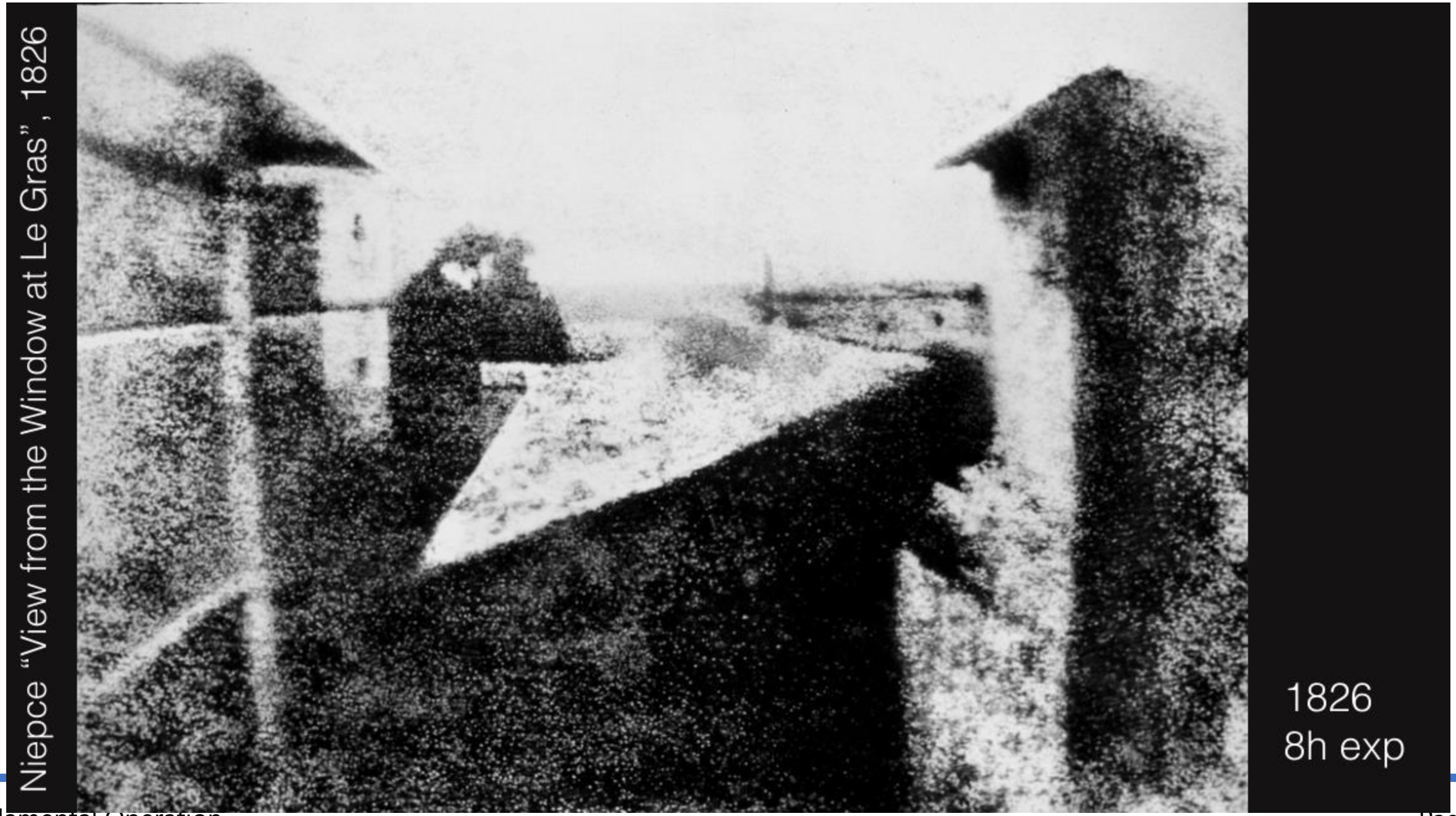
Camera obscura (the beginnings of photograph)





Ca




Niepce “View from the Window at Le Gras”, 1826
1826
8h exp



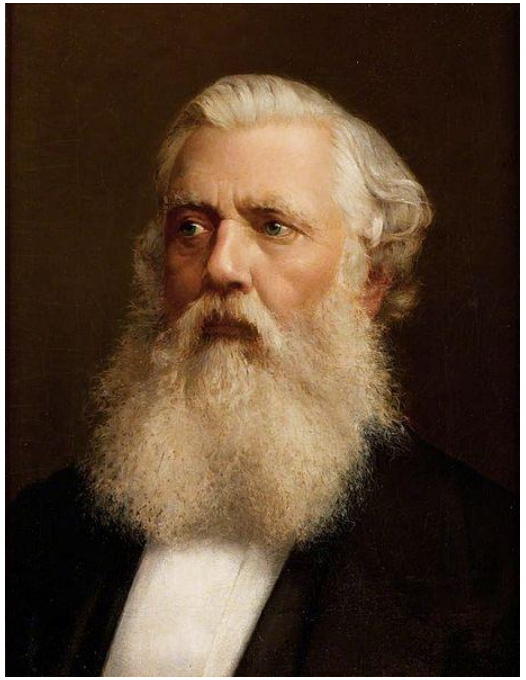
Louis Daguerre

Lens focus light
Better chemical ➔ 10 – 12 min

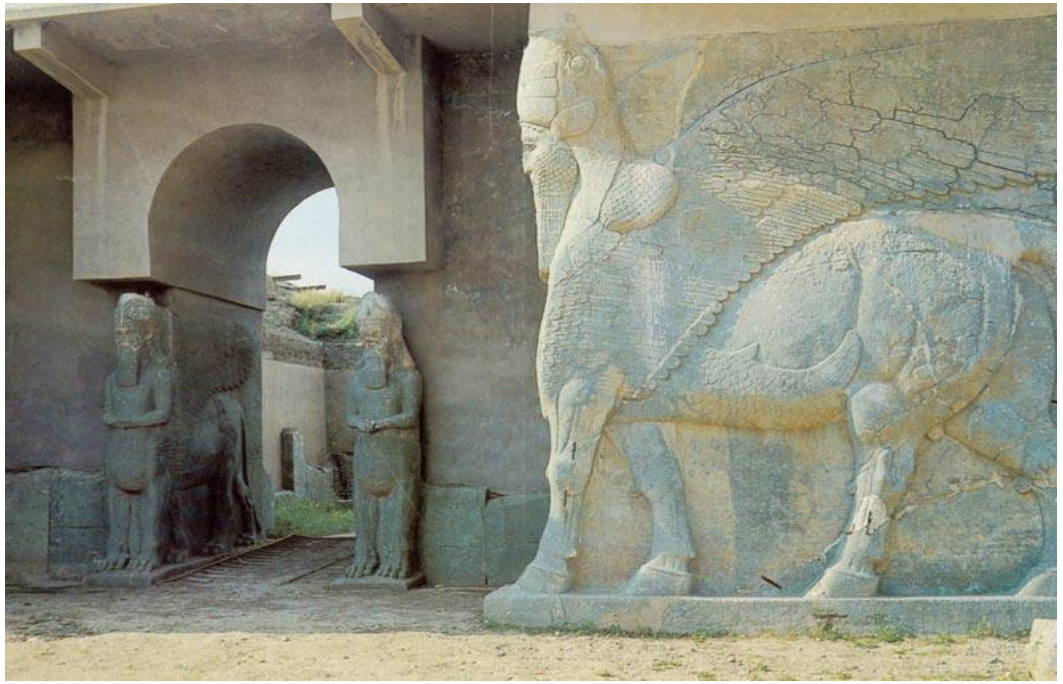




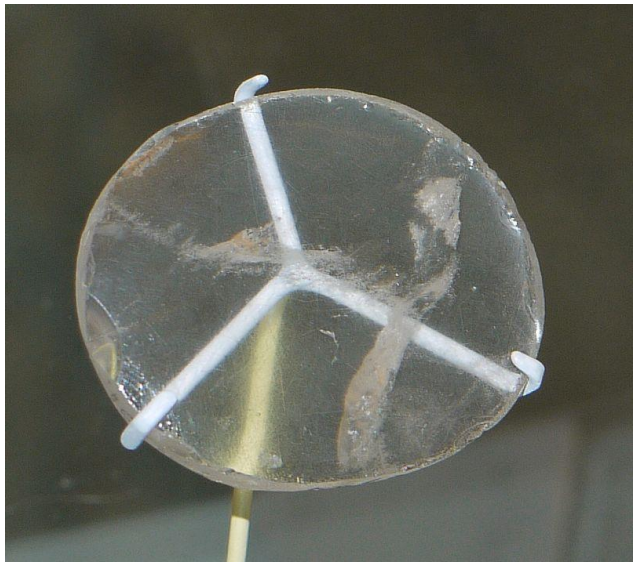
Lens (???)



Sir Austen Henry Layard
English traveller



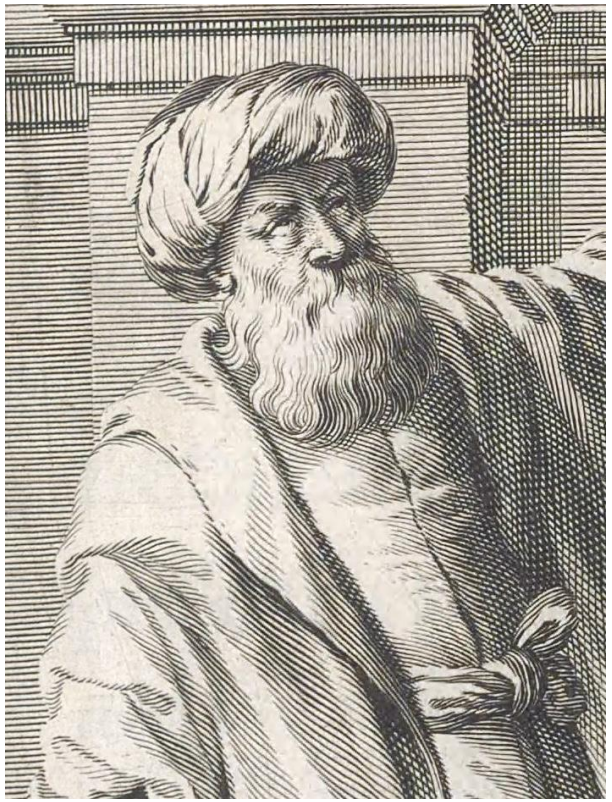
Assyrian palace reliefs
ancient Assyrian states (Iraq)



Nimrud lens

Alhazen
Ḥasan Ibn al-Haytham

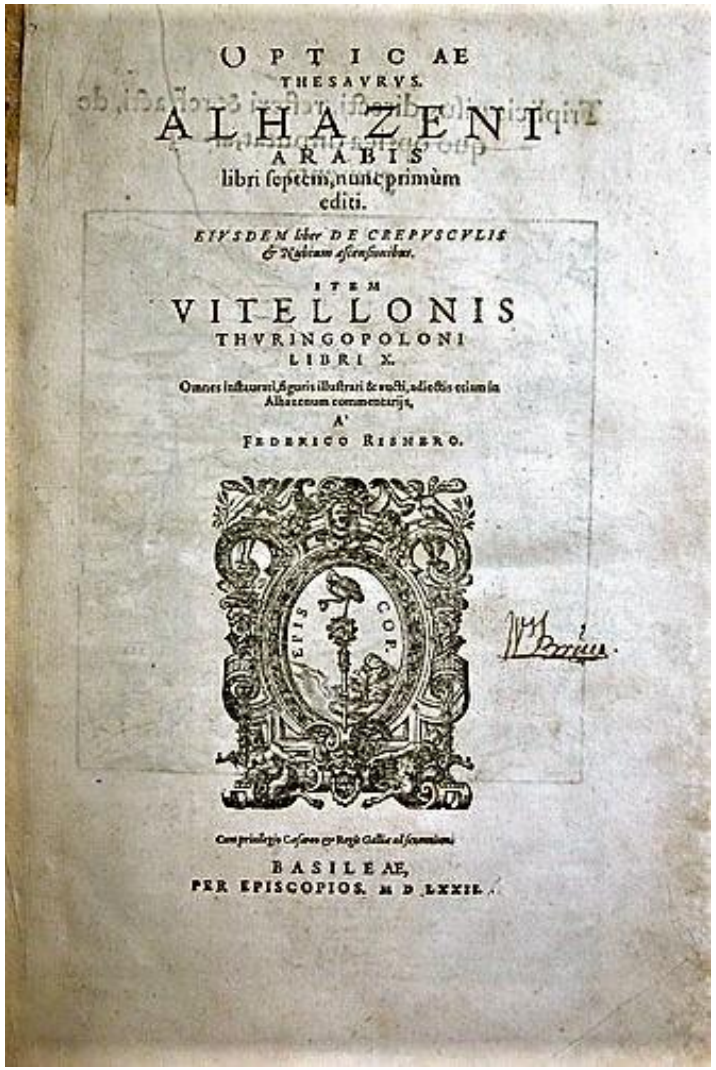
ابن الهيثم



อิบนุ ฮัยซัม (Ibn al-Haytham)

หรือที่รู้จักในชื่อละตินว่า Alhazen

- ยุค: ศตวรรษที่ 10-11 (ประมาณ ค.ศ. 965 – 1040)
- ผลงานเด่น: เขียนหนังสือชื่อ "Kitab al-Manazir" (Book of Optics)
- **สิ่งที่ค้นพบ**:
  - พิสูจน์ว่าแสงเดินทางเป็นเส้นตรง
  - อธิบายหลักการการสะท้อนและหักเหของแสง
  - ใช้เลนส์นูนเพื่อ **ขยายภาพ**
  - วางรากฐานของ กล้องถ่ายภาพ (Camera Obscura)



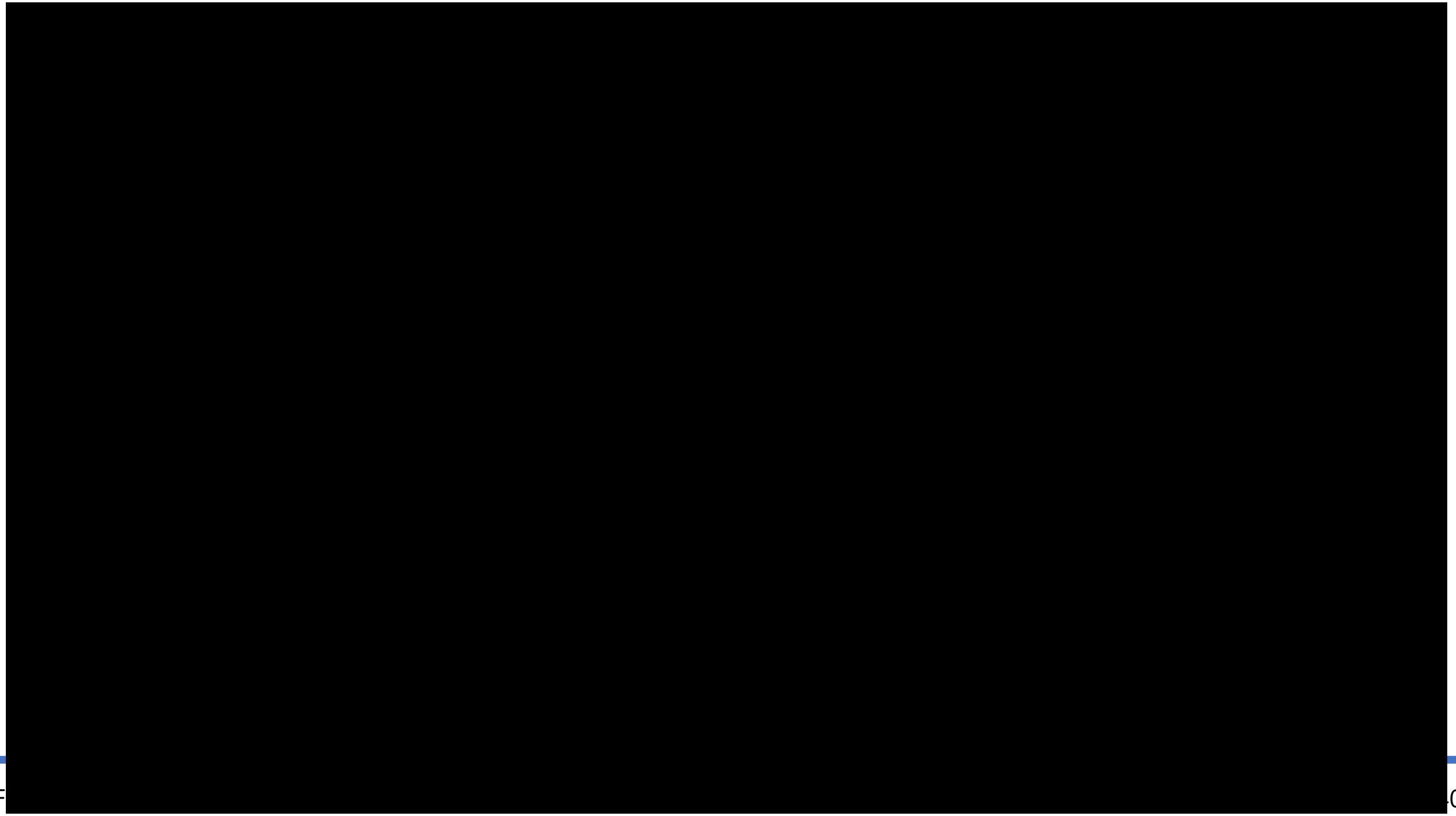


Image Acquisition (Pinhole Camera)

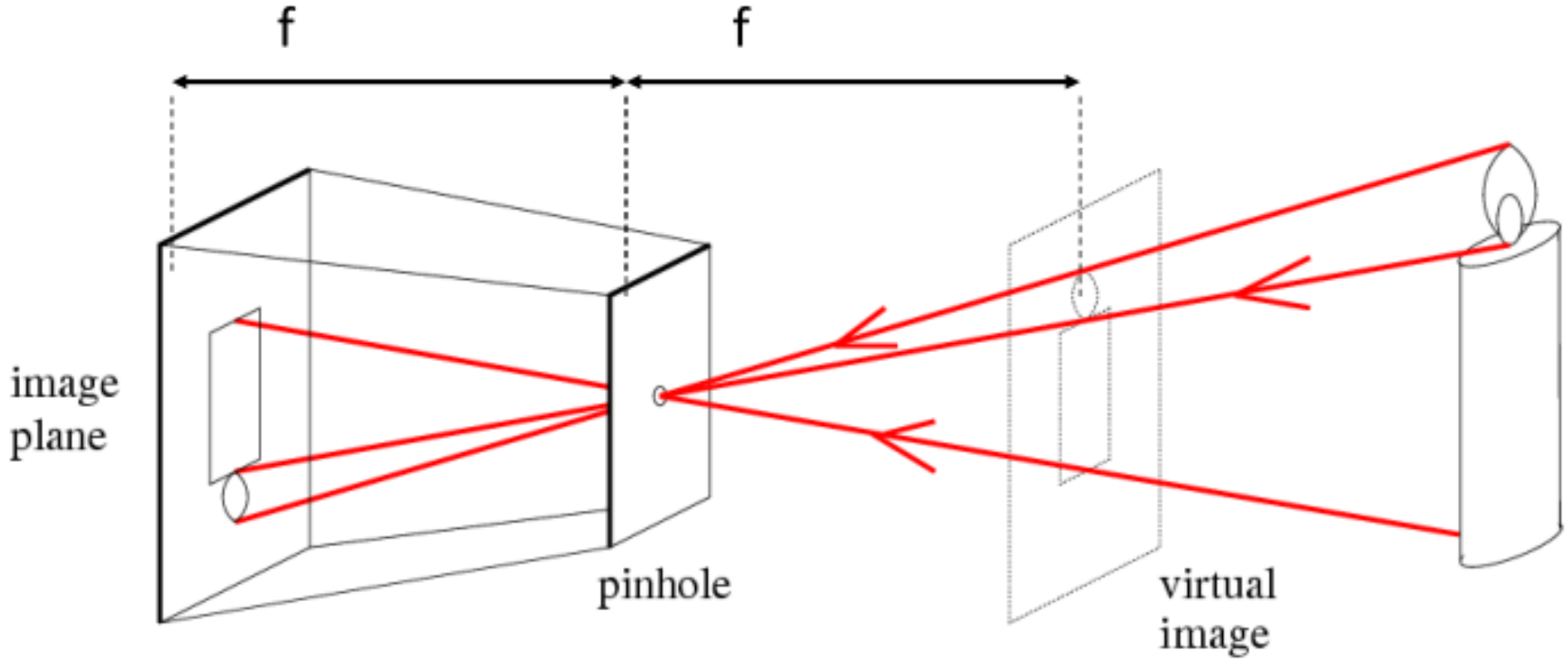
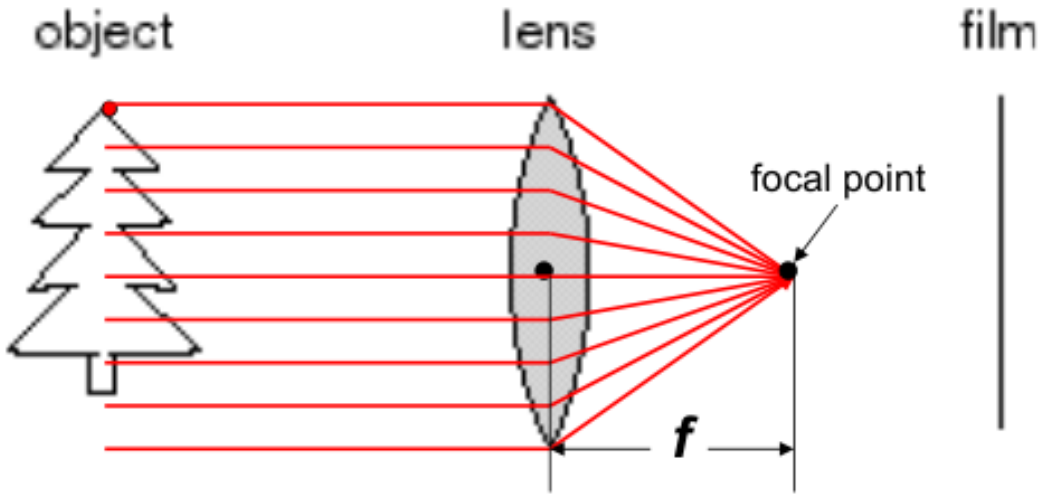


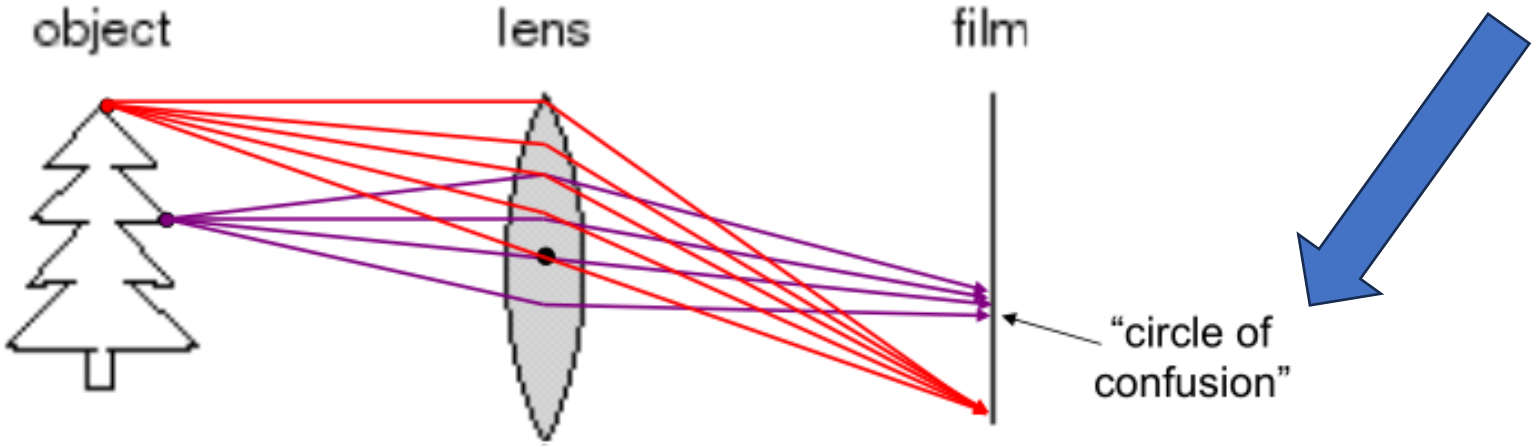
Image Acquisition (Lens)



A lens focuses light onto the film

- Rays passing through the center are not deviated
- All parallel rays converge to one point on a plane located at the *focal length f*

Image Acquisition (Lens)



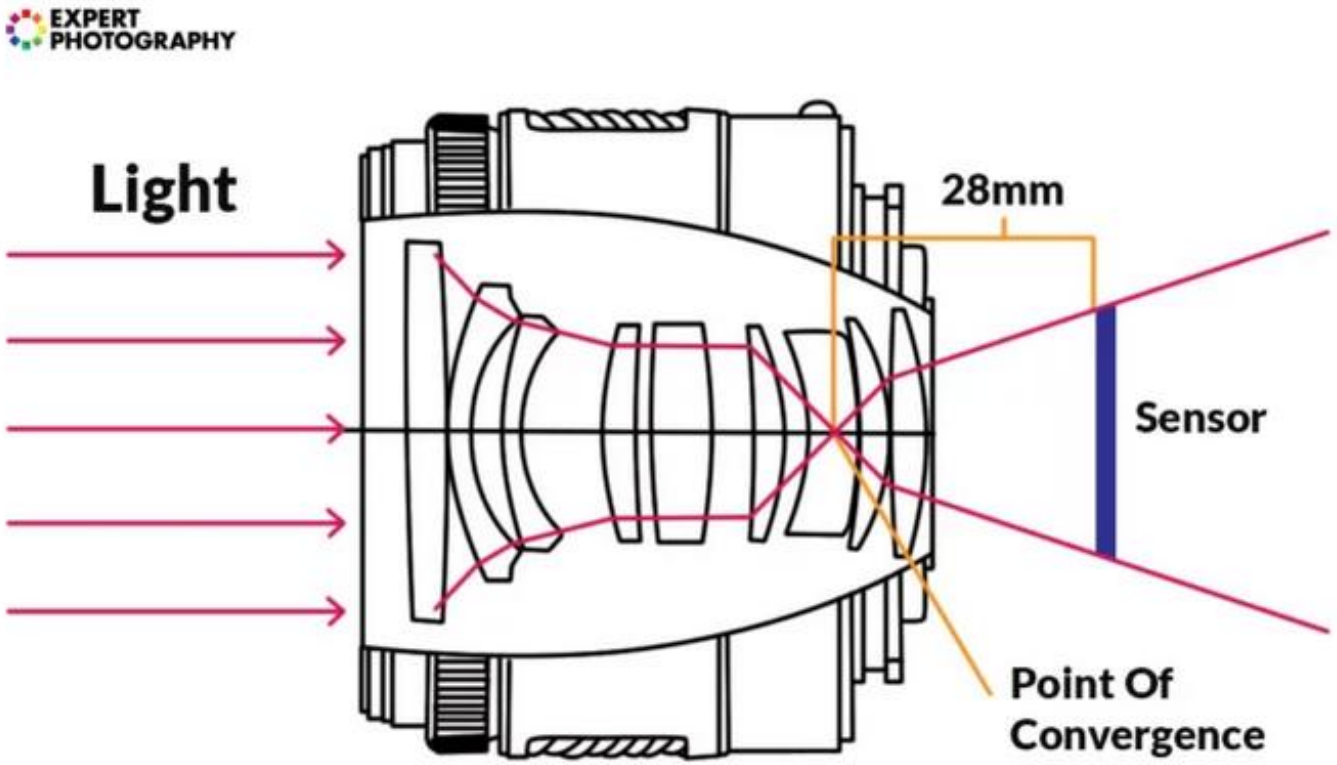
A lens focuses light onto the film

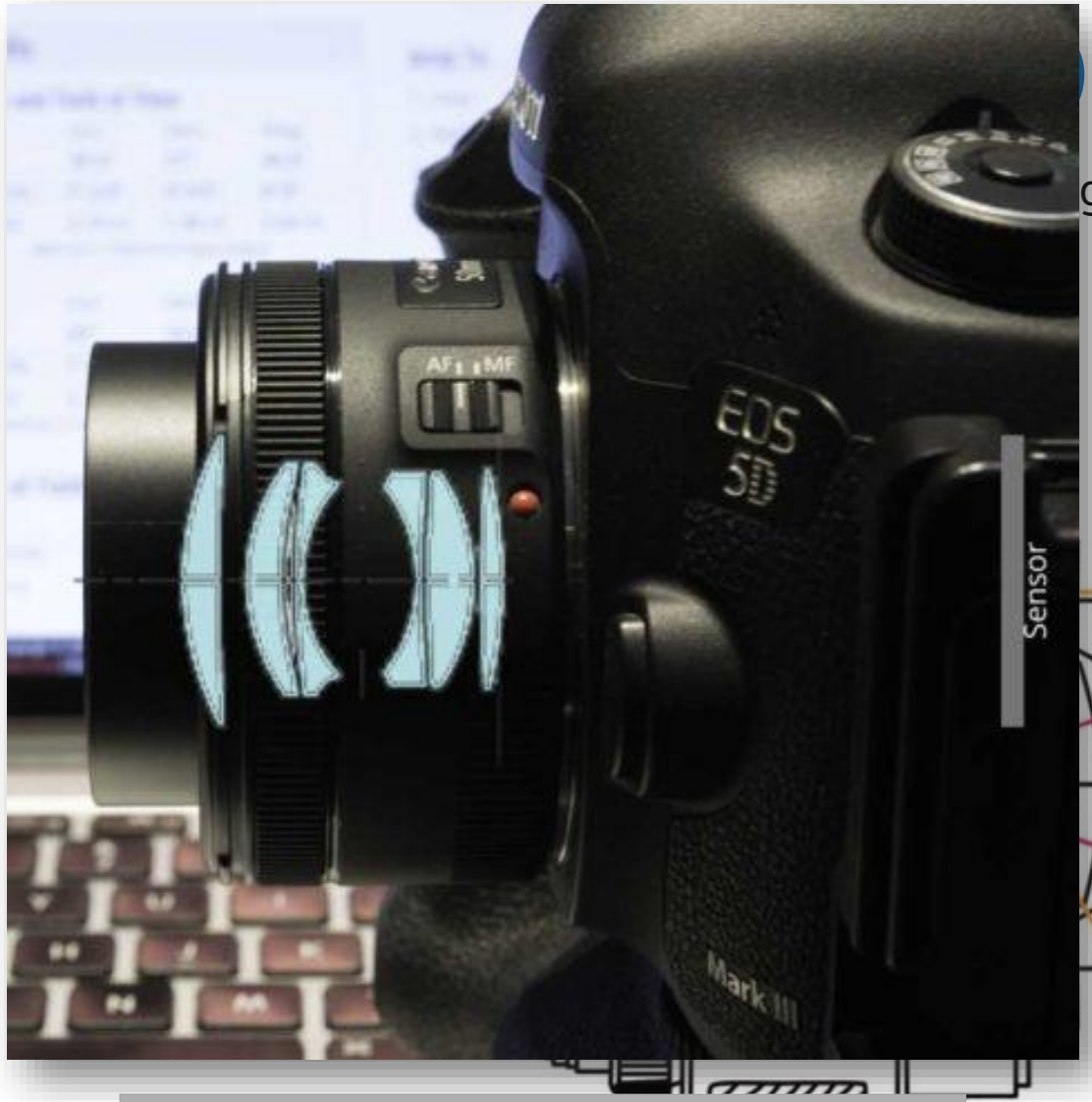
– There is a specific distance at which objects are "in focus"

[other points project to a "circle of confusion" in the image]

Image Acquisition (Lens)

Why do we use complex, multi-element designs and large lens housings if a single lens will also provide an image for much cheaper?





gns a

28m



Convergence

Real Glass

Approximate (Thin lens)

Image Acquisition (Lens)

Snell's law

$$n_1 \sin \alpha_1 = n_2 \sin \alpha_2$$

α_1 = incident angle
α_2 = refraction angle
n_i = index of refraction

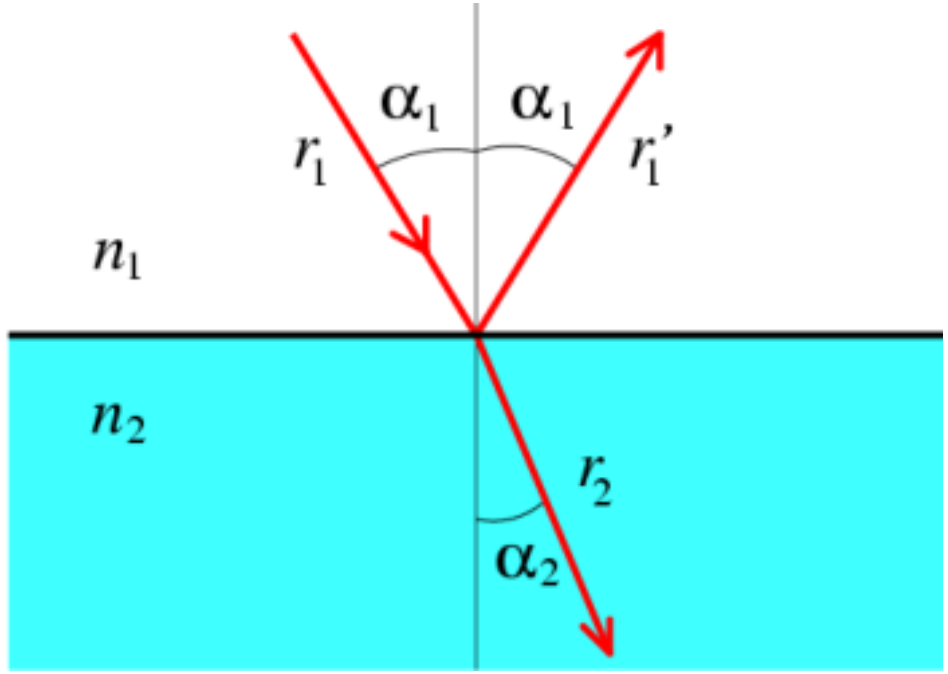
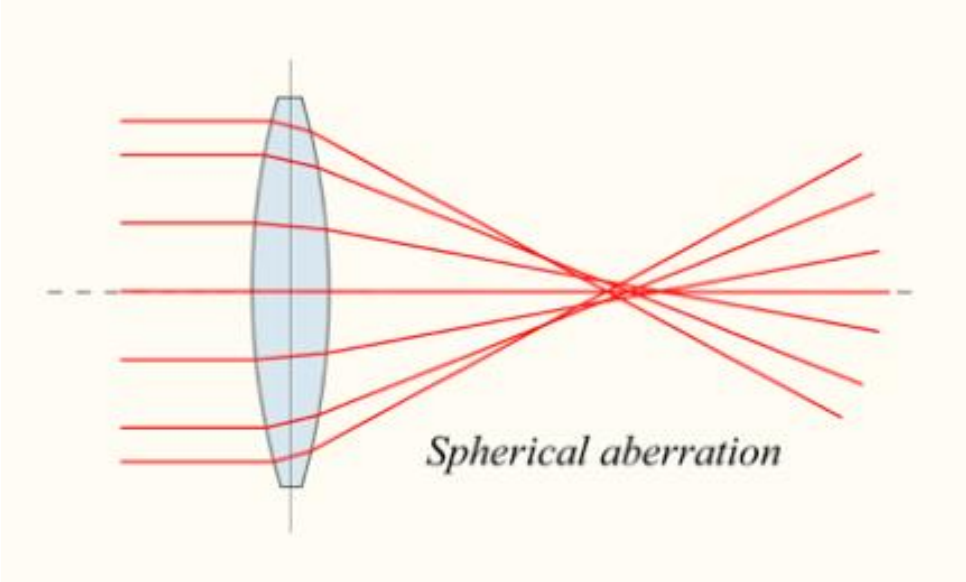
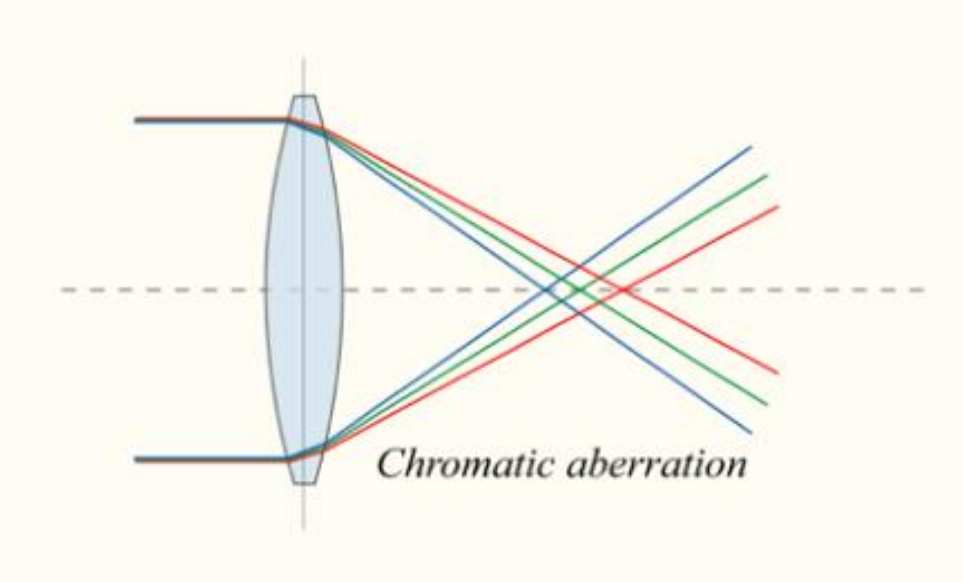


Image Acquisition (Aberrations (Abnormal))



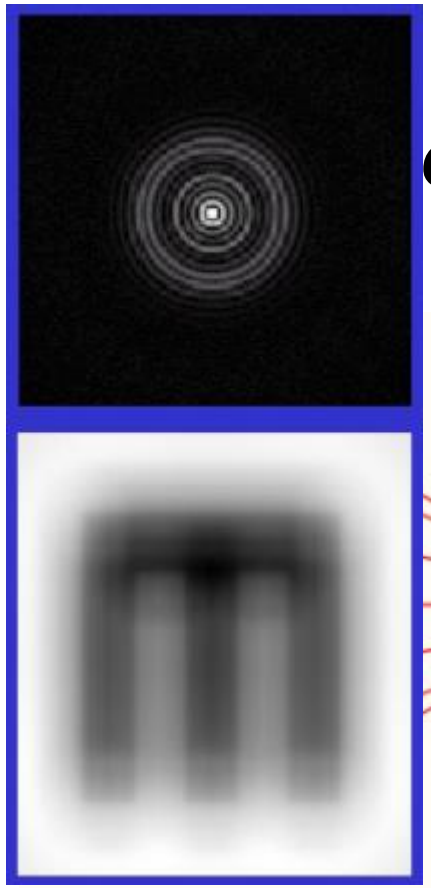
Rays farther from the optical axis focus closer

(แถวๆ ขอบเลนส์จะโฟกัสใกล้กว่า)

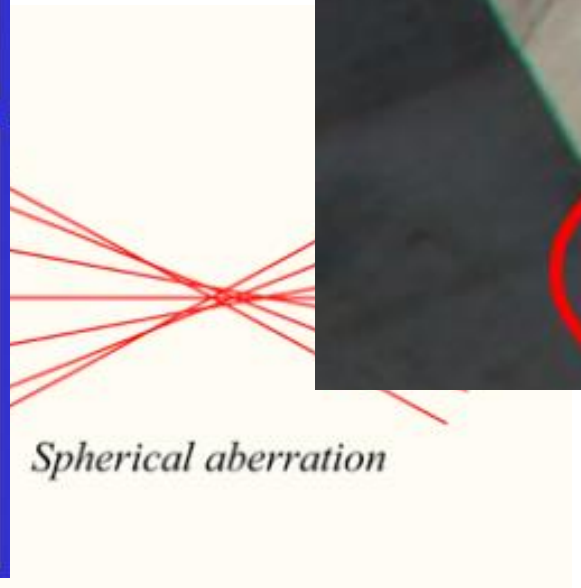


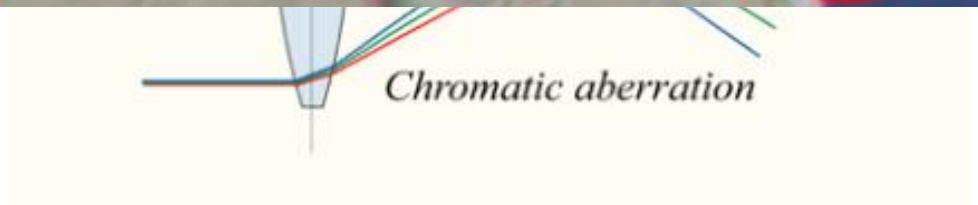
Lens has different refractive indices for different wavelengths: causes colour fringing

quisition (Aberrations (Abnormal))







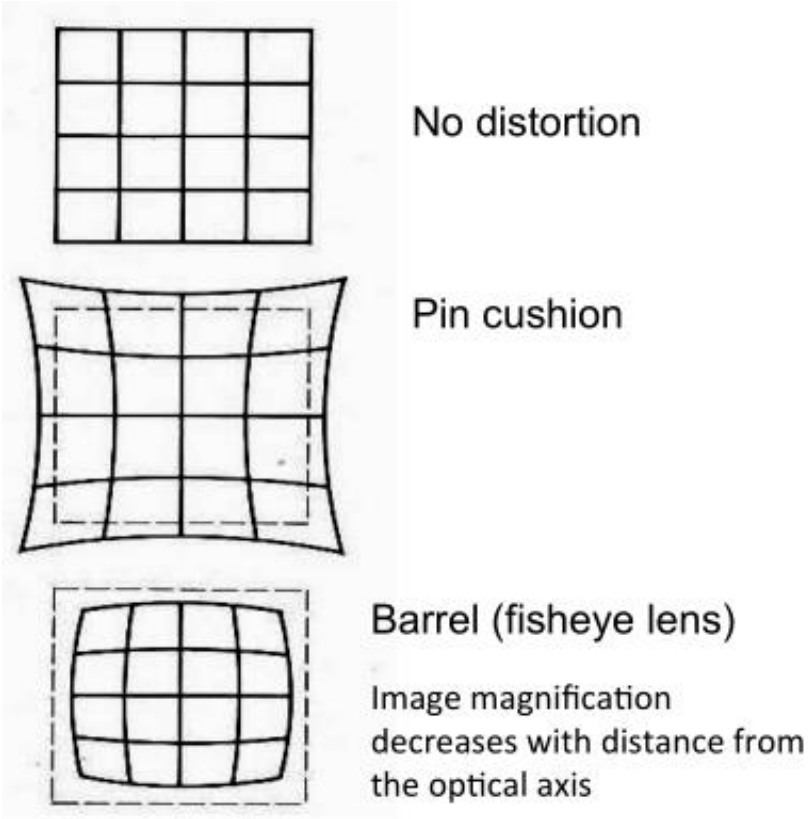


Rays farther from the optical axis focus closer

(แถวๆ ขอบเลนส์จะโฟกัสใกล้กว่า)

Lens has different refractive indices for different wavelengths: causes colour fringing

Image Acquisition (Aberrations (Abnormal))





Deviations are most noticeable for rays that pass through the edge of the lens

How to choose camera?
(for image processing project)

[1] **Data type** (Still Image or Video) ➔ **CCD** or **CMOS**

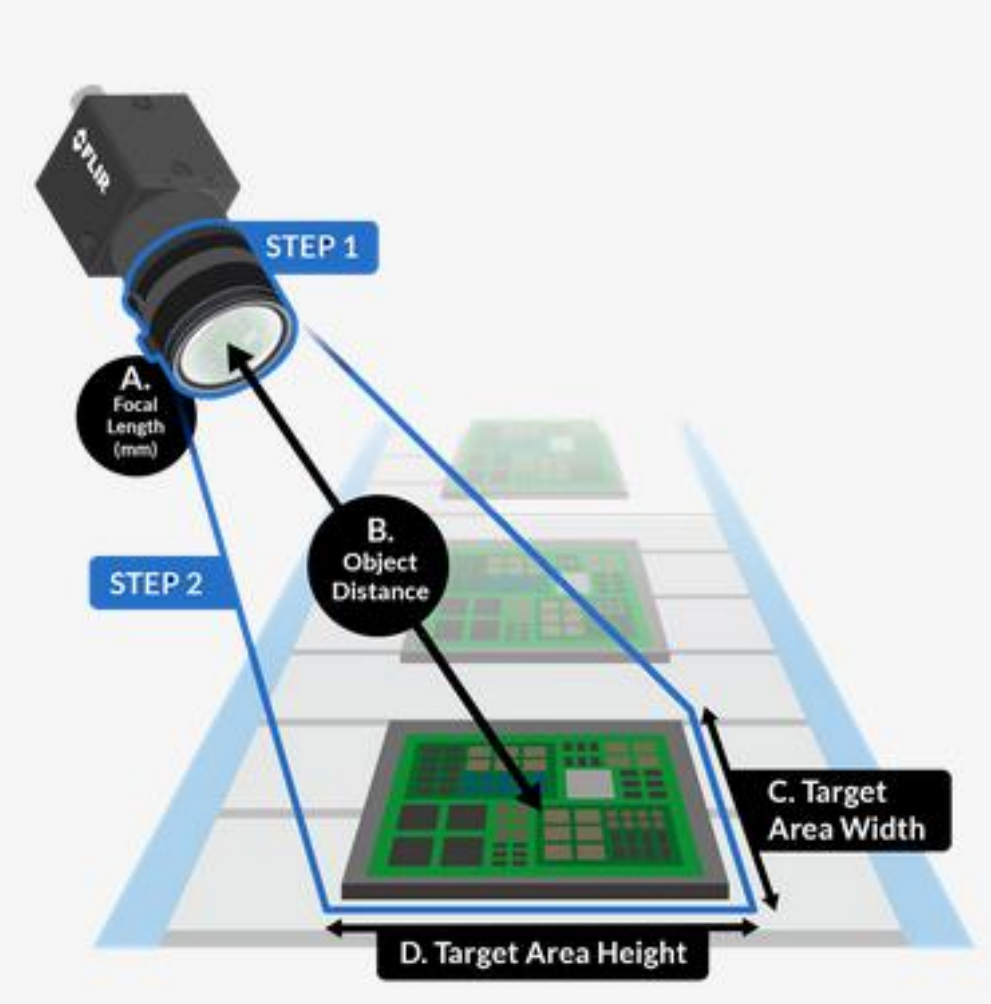
[2] **Coverage area** (**Field of View** / **Size of Object**)

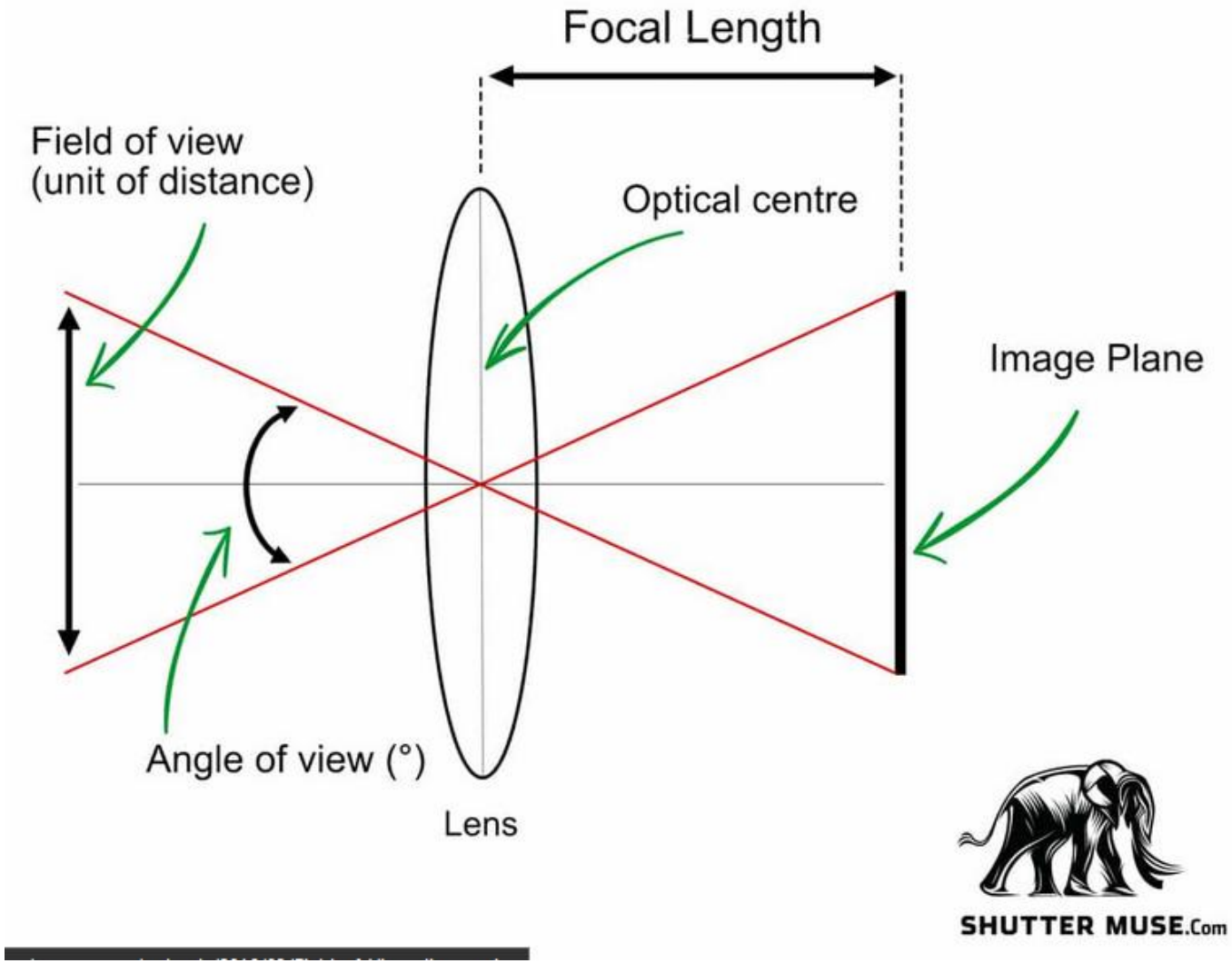
[3] **Resolution of image** / **Frame rate** (**Detection** / **Identify**)

[4] **Connectivity** / **Hardware Memory**

[5] **Reasonable price**

Field of View (FOV)

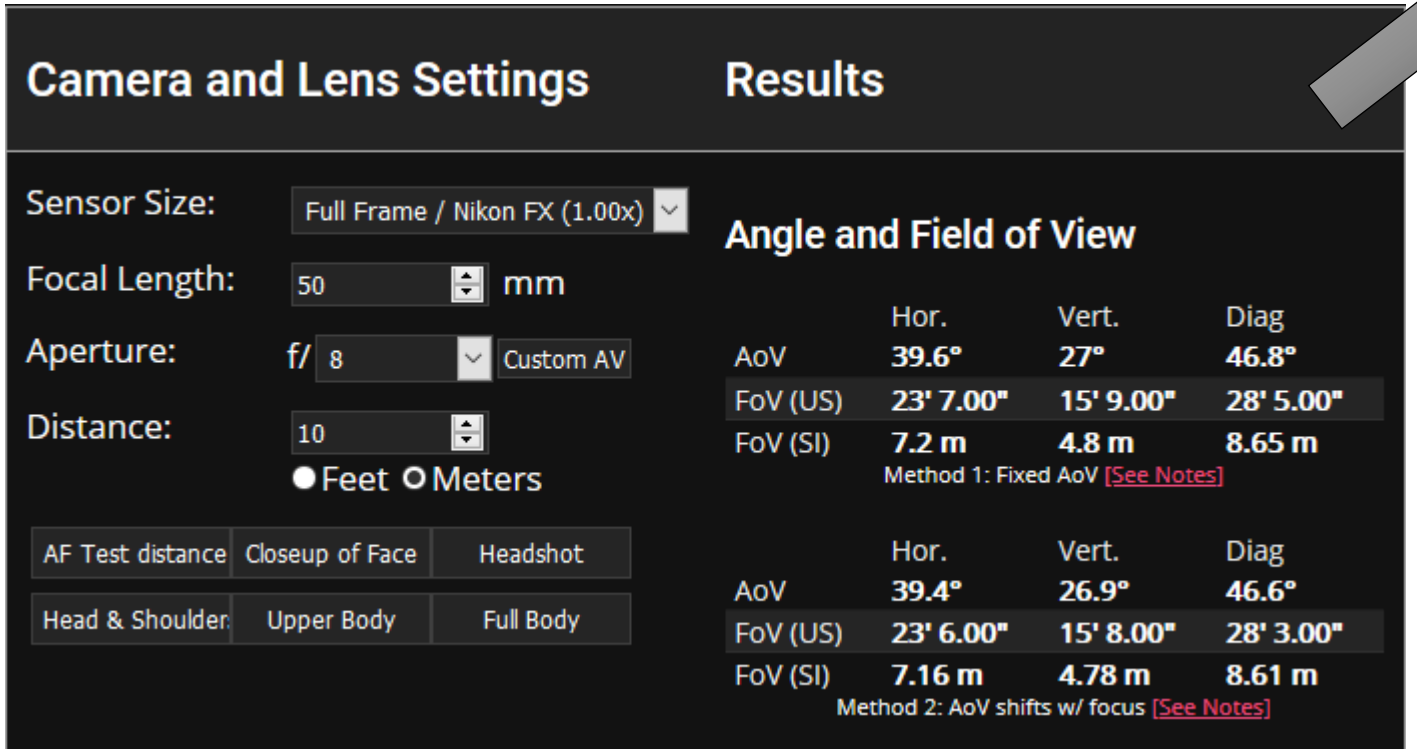




SHUTTER MUSE.Com

Check your camera

Recommend to do it with calculator software such as Microsoft Excel, etc,.



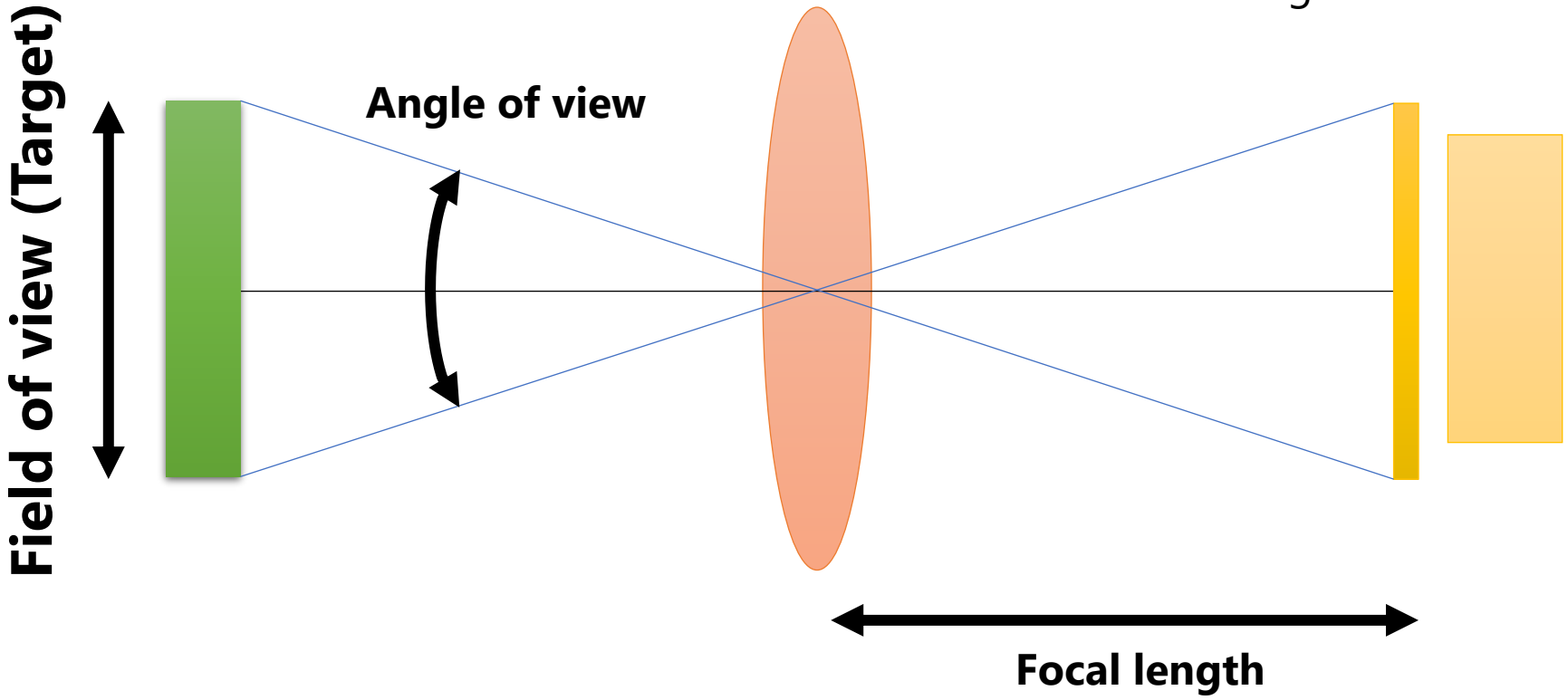


https://www.pointsinfocus.com/tools/depth-of-field-and-equivalent-lens-calculator/

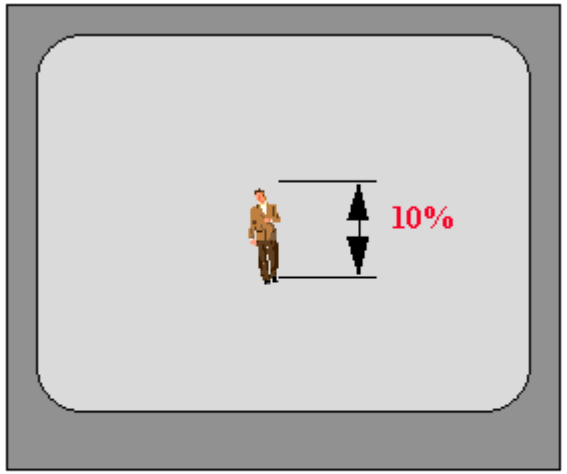
Check your camera

Find "**Angle of view**"
from Focal length & Sensor size

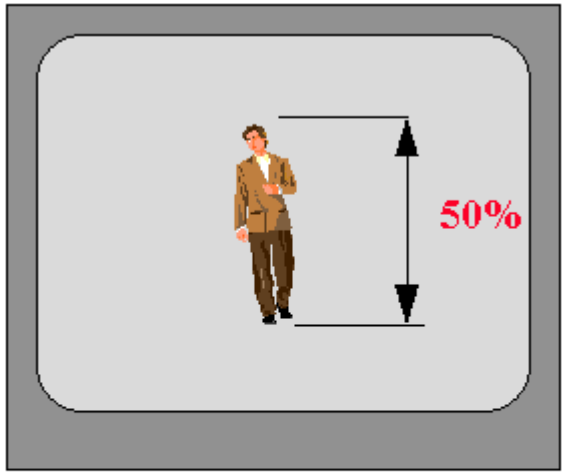
Find "**Field of view**"
From Angle of view & Distance to Obj.



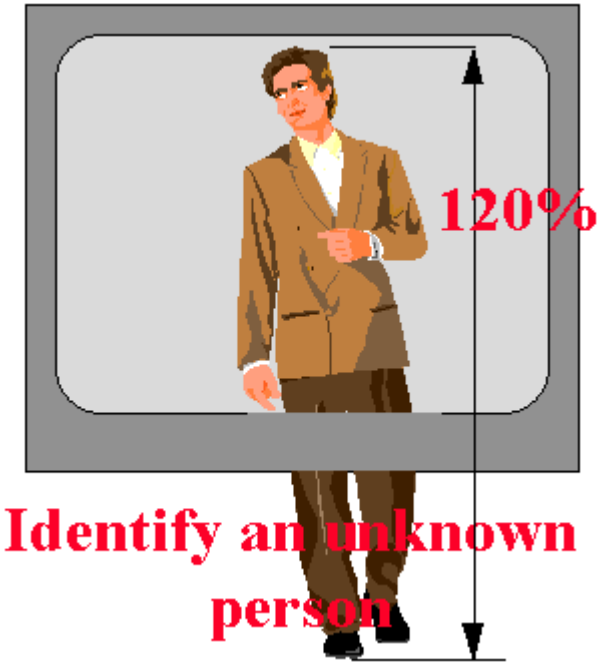
Choose Sensor & Focal Length with your **GOAL**



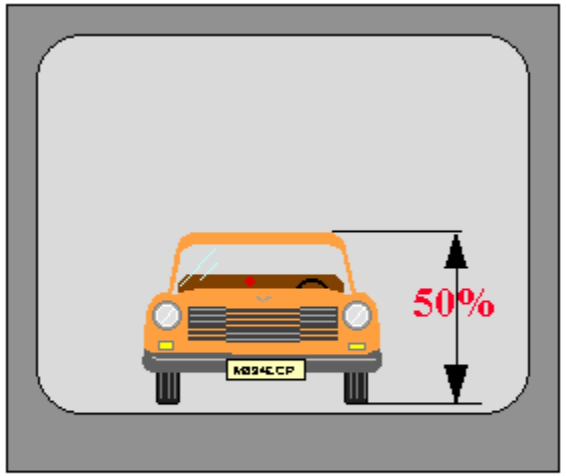
Detect a person



Recognise a known person



Identify an unknown person



Car Number Plate

เรื่องการคำนวณ Field of view (FOV)

[1] หา Angle of view ($\boldsymbol{\theta}$) จาก sensor size กับ Focal length ($\boldsymbol{f}$)

$$\theta = 2\tan^{-1}\left(\frac{\text{sensor size}}{2f}\right)$$

[2] หา Field of view (FOV) จาก Angle of view กับ Distance to Object ($\boldsymbol{S_o}$)

$$\text{FOV} = 2\left(S_o * \tan\left(\frac{\theta}{2}\right)\right)$$

เรื่องการคำนวณ Field of view (FOV)

ข้อแนะนำ: ควรระมัดระวังเรื่องการใช้หน่วยวัด (ให้ใช้หน่วยวัดเดียวกันจะได้ไม่สับสน)

$$\theta = 2\tan^{-1}\left(\frac{\text{sensor size}}{2f}\right) \Rightarrow \theta = 2\tan^{-1}\left(\frac{35\text{mm}}{2(50mm)}\right)$$

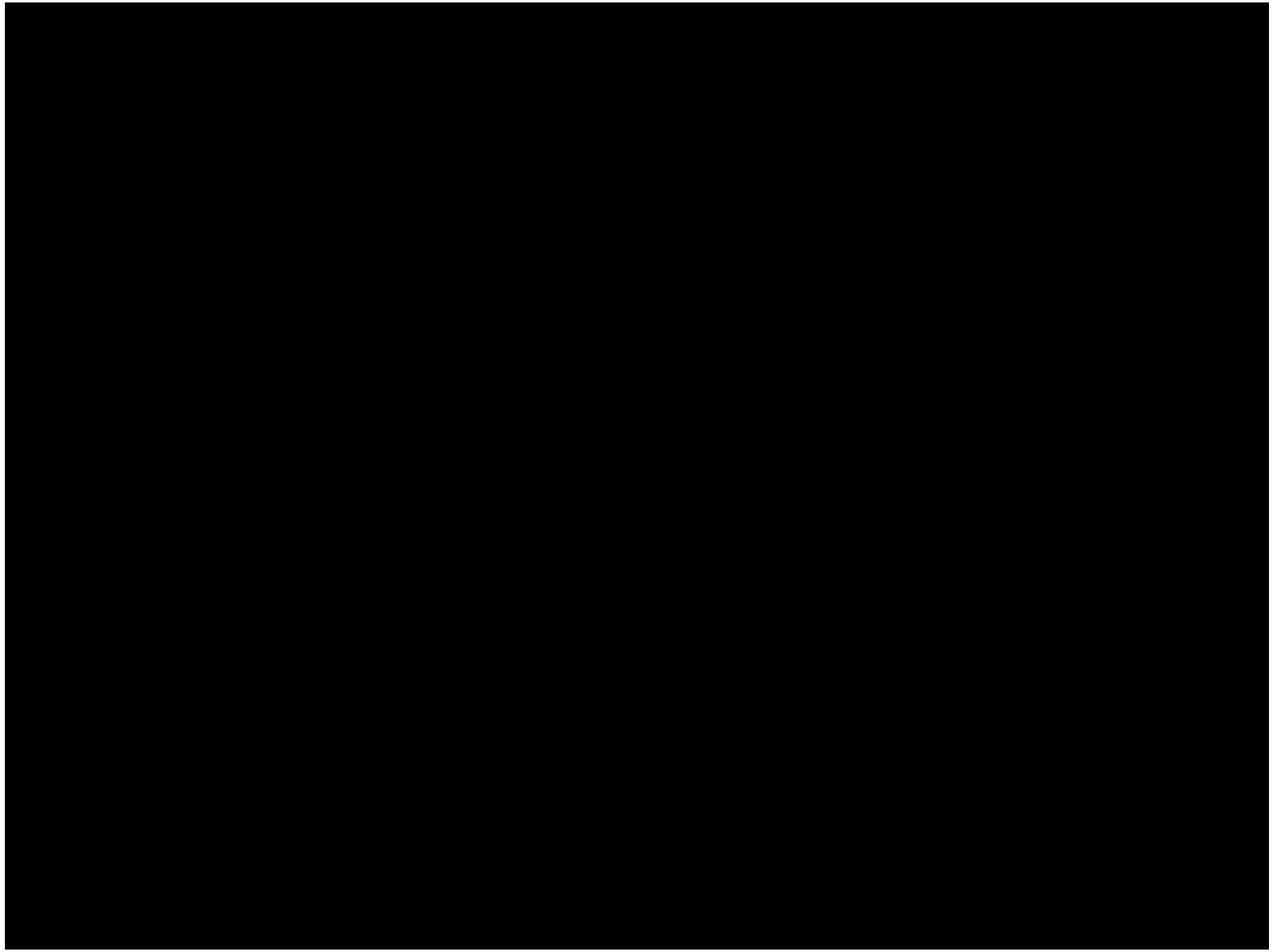
$$\text{FOV} = 2\left(S_o * \tan\left(\frac{\theta}{2}\right)\right) \Rightarrow \text{FOV} = 2\left((5m) * \tan\left(\frac{\theta}{2}\right)\right)$$

Field of view (FOV) จะได้ผลลัพธ์เป็นหน่วย เมตร (m) ตามหน่วยของระยะห่างวัตถุถึงเลนส์

Sensor

	CCD	CMOS
Resolution	Up to 100+ Megapixels	Up to 100+ Megapixels
Frame rate	Low frame rates	High frame rates
Noise figure	Low noise floor	High noise floor
Sensitive of intensity	More sensitive at low intensity	Less sensitive at low intensity
Shutter type	Global	Rolling
Skew	No	Yes

Sensor



Classwork

- ตรวจสอบคุณสมบัติของกล้องของคุณ
  - Focal Length
  - Sensor Size
  - Field of View
  - Resolution (Pixel per Inch)
- ในระยะ 1 เมตร มี Field of View ขนาดเท่าใด (กว้าง x ยาว)

  ตรวจสอบหน่วยวัดด้วย (cm หรือ m)
- ภาพที่ถ่ายโดยกล้องของคุณ มี Distortion เกิดขึ้นหรือไม่ (ตรวจสอบเส้นตรง)
- ค่าสีที่ได้จากกล้องของคุณ มีความแตกต่างจากเพื่อนในกลุ่มคนอื่นหรือไม่
- ทำไฟล์สรุป 1 กลุ่ม ส่ง 1 คน (.pdf)